

Trade Tariffs and Exchange Rates in a Multilateral World

Theoretical derivations

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September 17, 2026

1 Introduction

In a two-country model, an import tariff appreciates the tariffing country's currency under the Marshall–Lerner condition. With more than two countries, a tariff on one trading partner also reallocates expenditure across foreign suppliers, so bilateral exchange rates need not all move in the same direction.

These notes characterize the exchange-rate effects of tariffs in an N -country model. The general log-linearized equilibrium response is $(-J)^{-1}F$, where F gives the trade-balance effects of a tariff at unchanged exchange rates and J is the Jacobian of trade balances with respect to exchange rates. Under a multilateral Marshall–Lerner (MML) condition on J , this representation separates the trade-balance effects of the tariff from the exchange-rate adjustment required to restore equilibrium.

Under symmetry, the response to any tariff vector further separates into an average-protection component and a relative-treatment component. Average protection determines the tariffing country's nominal effective exchange rate (NEER), while relative treatment determines the dispersion of bilateral exchange-rate responses around it. For a unilateral tariff, the tariffing country always appreciates against the targeted partner and its NEER always appreciates, but its exchange rate against an untariffed bystander can move in either direction. A bystander depreciation of the tariffing country's currency occurs only when substitution across foreign import origins is sufficiently strong relative to the other trade-balance effects of the tariff (Section 4.2). Proofs are collected in Appendix G.

2 Tariffs and Exchange-Rate Adjustment

The results in this section require only that trade balances are differentiable with respect to exchange rates and tariffs, and do not depend on our specific nested-CES model structure.

2.1 The bilateral case

Consider first two countries, A and B . Let e_{AB} be units of A 's currency per unit of B 's, so that a rise in e_{AB} is a depreciation of A , and write $\nu \equiv d \log e_{AB}$. Let A 's trade balance TB_A be a differentiable function of ν and of a tariff τ that A levies on B . Linearizing at a balanced free-trade

baseline,

$$dTB_A = TB_{A\nu} \nu + TB_{A\tau} d\tau, \quad TB_{A\nu} \equiv \frac{\partial dTB_A}{\partial \nu}, \quad TB_{A\tau} \equiv \frac{\partial dTB_A}{\partial \tau} \Big|_{\nu \text{ fixed}}. \quad (1)$$

The tariff raises the domestic price of imports and so improves A 's trade balance at unchanged exchange rates, $TB_{A\tau} > 0$. Restoring balance then requires

$$\nu^* = -\frac{TB_{A\tau}}{TB_{A\nu}} d\tau, \quad (2)$$

which is an appreciation if and only if $TB_{A\nu} > 0$, that is, if and only if a depreciation improves A 's trade balance. This is the ordinary Marshall–Lerner condition; Section 4.1 shows that in the nested-CES model it takes the familiar elasticity form $\varepsilon_X + \varepsilon_M > 1$.

The two steps in (1)–(2) are the whole of the adjustment logic: the tariff first displaces the trade balance at unchanged exchange rates, and the exchange rate then moves to absorb that displacement. Figure 1 displays them.

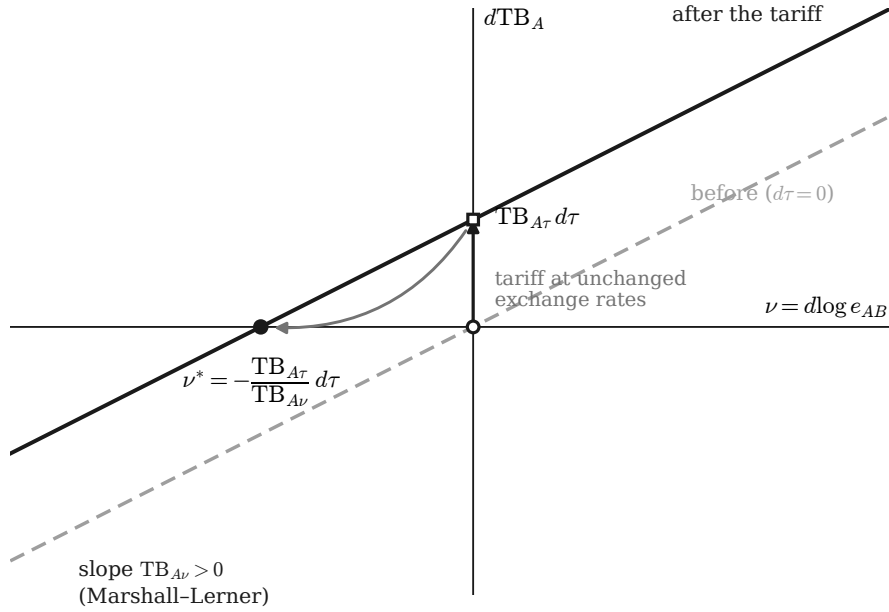


Figure 1: Bilateral equilibrium adjustment. The schedule is (1); its slope $TB_{A\nu}$ is positive exactly under the Marshall–Lerner condition. At unchanged exchange rates the tariff shifts the schedule up by $TB_{A\tau} d\tau > 0$ (open square). The currency then appreciates along the new schedule until balance is restored at $\nu^* = -(TB_{A\tau}/TB_{A\nu}) d\tau$.

2.2 The local disturbance–adjustment representation

Consider any differentiable multilateral trade environment with $N - 1$ independent trade-balance conditions, $TB_i = 0$ for $i \neq A$, which are each functions of exchange rates and tariffs. Stack the exchange-rate changes as $\nu \equiv (\nu_j)_{j \neq A}$, $\nu_j \equiv d \log e_{Aj}$, and the bilateral tariff changes as

$d\tau \equiv (dt_{jk})_{j \neq k}$, so that ν and $d\tau$ are the perturbation vectors of the levels e_{Aj} and τ_{jk} . The conditions for $i \neq A$ are

$$J\nu + Gd\tau = 0, \quad J_{il} \equiv \frac{\partial dTB_i}{\partial \nu_l}, \quad G_{i,(jk)} \equiv \frac{\partial dTB_i}{\partial t_{jk}} \Big|_{\nu \text{ fixed}}. \quad (3)$$

A tariff experiment fixes a *pattern* of tariff changes a and a scalar *magnitude* $d\tau$, so that $d\tau = a d\tau$. Defining

$$F \equiv Ga, \quad (4)$$

the vector of fixed-exchange-rate trade-balance disturbances generated by one unit of that pattern, the equilibrium conditions become

$$J\nu = -F d\tau, \quad \nu = (-J)^{-1} F d\tau. \quad (5)$$

Thus F , and more generally G , gives the trade-balance effects of the tariff at unchanged exchange rates, including substitution and income effects; J describes how trade balances respond to exchange-rate changes; and $(-J)^{-1}$ maps the disturbance into the equilibrium exchange-rate changes that absorb it.

At $N = 2$ there is a single condition, $TB_B = 0$. Since $dTB_B = -dTB_A$, the scalars in (3) are $-J = TB_{A\nu}$ and $F = -TB_{A\tau}$, and (5) is exactly (2). The N -country system is the matrix analogue of that scalar problem.

2.3 The multilateral Marshall–Lerner condition

At $N = 2$, the requirement that exchange-rate adjustment stabilizes the trade balance is the scalar condition $-J = TB_{A\nu} > 0$ of Section 2.1. The N -country generalization requires that the adjustment mapping $(-J)^{-1}$ be well defined and positive.

Definition 2.1 (Multilateral Marshall–Lerner condition). *The adjustment mechanism satisfies the multilateral Marshall–Lerner condition (MML) if*

$$-J \text{ is a nonsingular } M\text{-matrix:} \quad J_{il} \geq 0 \quad (l \neq i) \quad \text{and} \quad (-J)^{-1} \text{ exists with } (-J)^{-1} \geq 0. \quad (6)$$

By (5), under MML, each exchange-rate response is a nonnegative-weighted combination of the trade-balance effects at fixed exchange rates. Writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{d\tau} = \sum_j \vartheta_{ij} F_j, \quad (7)$$

so ϑ_{ij} measures the response of currency i to a unit trade-balance disturbance in country j . A uniformly signed F therefore cannot generate exchange-rate responses of the opposite sign. When F has mixed signs, the response of any individual exchange rate depends on the full vector of trade-balance effects rather than on the corresponding element F_i alone. The condition also implies

$\det(-J) > 0$, which is used in the threshold calculations below. At $N = 2$, the system is scalar and the condition reduces to $TB_{A\nu} > 0$, the ordinary Marshall–Lerner condition.

The condition above is stated directly on J , but it can alternatively be obtained from a restriction on the full bilateral Jacobian. Let $\tilde{J}$ be the full $N \times N$ Jacobian, so that J is $\tilde{J}$ with row and column A deleted. Two identities hold for $\tilde{J}$. Each trade balance is homogeneous of degree one in currency values, so by Euler’s theorem row i sums to TB_i , which is zero at a balanced baseline; and Walras’ law, $\sum_i TB_i \equiv 0$, ensures that the columns sum to zero. Thus

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i, \quad \sum_i \tilde{J}_{il} = 0 \quad \forall l. \quad (8)$$

The row identity pins down each diagonal entry of $\tilde{J}$ from its off-diagonal entries, so a sign restriction on the off-diagonal entries alone determines the own effects and, after deleting A , the dominance of J . This gives a sufficient condition for MML.

Assumption 1 (Gross substitutability). $\tilde{J}_{il} \geq 0$ for all $l \neq i$.

An appreciation of currency l raises the price of l ’s goods, and Assumption 1 says that this improves every other country’s trade balance: tradables are gross substitutes across countries. With the row identity in (8), $\tilde{J}_{ii} = -\sum_{l \neq i} \tilde{J}_{il} \leq 0$, and deleting row and column A gives

$$|J_{ii}| - \sum_{l \neq i, A} J_{il} = \tilde{J}_{iA} \geq 0, \quad (9)$$

so J is diagonally dominant, strictly in every row with $\tilde{J}_{iA} > 0$.

Lemma 2.2 (Sufficiency of gross substitutability). *Under Assumption 1, with J irreducible and $\tilde{J}_{iA} > 0$ for some i , MML (6) holds.*

The proof is in Appendix G, which also records the equivalent spectral characterization and its implication for tatonnement stability.

2.4 Symmetric partners

Restrict attention to tariffs imposed by A , and write $t = (t_j)_{j \neq A}$ with $t_j \equiv dt_{Aj}$ for A ’s own tariff vector, so that G in (3) is $(N - 1) \times (N - 1)$. Suppose that A ’s trading partners are symmetric: relabeling them leaves the reduced system unchanged, $PJP' = J$ and $PGP' = G$ for every permutation matrix P of the $N - 1$ partners. This holds when the partners have identical sizes, trade weights, and parameters at the baseline; A itself need not be symmetric with them. Take $N \geq 3$.

Under partner symmetry, the tariff vector can affect exchange rates through only two components: average protection, the average tariff across partners, and relative treatment, the differences in tariffs across partners. Decompose

$$t = \bar{t} \mathbf{1} + \tilde{t}, \quad \bar{t} \equiv \frac{1}{N-1} \mathbf{1}' t, \quad \mathbf{1}' \tilde{t} = 0. \quad (10)$$

Componentwise this is

$$t_j = \bar{t} + (t_j - \bar{t}), \quad \tilde{t}_j \equiv t_j - \bar{t}. \quad (11)$$

We refer to $\bar{t}\mathbf{1}$ as *average protection* and to $\tilde{t}$ as *relative treatment*: average protection is the tariff level A applies to its partners on average, and relative treatment records how each partner is taxed relative to that average. Mathematically, $\mathbf{1}$ spans the *scale* eigenspace of constant vectors and $\tilde{t}$ lies in the *discriminatory* eigenspace of zero-sum vectors; we use the latter terms only where the eigenspace structure itself is at issue.

Let $d\nu_A^E \equiv \frac{1}{N-1} \sum_{j \neq A} \nu_j$ denote the equally weighted average of A 's bilateral responses, which under symmetry is A 's nominal effective exchange rate (21).

Proposition 2.3 (Average protection and relative treatment). *Suppose A 's partners are symmetric and $N \geq 3$. Then $-J$ and G each have exactly two eigenvalues, one on $\text{span}\{\mathbf{1}\}$ and one on $\{v : \mathbf{1}'v = 0\}$; write these j_S, j_D and g_S, g_D respectively. If $j_S, j_D \neq 0$, the response to any tariff vector t imposed by A is*

$$\nu = \lambda_S \bar{t} \mathbf{1} + \lambda_D \tilde{t}, \quad \lambda_S \equiv \frac{g_S}{j_S}, \quad \lambda_D \equiv \frac{g_D}{j_D}, \quad (12)$$

so that average protection alone determines the effective exchange rate and relative treatment alone determines the dispersion of bilateral rates around it and the cross rates among partners:

$$d\nu_A^E = \lambda_S \bar{t}, \quad \nu_j - d\nu_A^E = \lambda_D \tilde{t}_j, \quad d \log e_{jk} = \lambda_D (t_k - t_j). \quad (13)$$

MML (6) holds if and only if

$$j_S > 0 \quad \text{and} \quad j_D \geq j_S. \quad (14)$$

The reason the response takes this two-term form is that any matrix invariant to relabeling the partners has the form $aI + b\mathbf{1}\mathbf{1}'$. With $\mathbf{1}'\mathbf{1} = N - 1$ and $\mathbf{1}'\tilde{t} = 0$,

$$(aI + b\mathbf{1}\mathbf{1}')t = (a + (N - 1)b)\bar{t}\mathbf{1} + a\tilde{t}.$$

The matrix therefore acts separately on average protection and relative treatment, with eigenvalues $a + (N - 1)b$ on $\text{span}\{\mathbf{1}\}$ and a on the zero-sum subspace. For $-J$ and G these are

$$\begin{aligned} -J = a_J I + b_J \mathbf{1}\mathbf{1}' : \quad j_S &= a_J + (N - 1)b_J, \quad j_D = a_J, \\ G = a_G I + b_G \mathbf{1}\mathbf{1}' : \quad g_S &= a_G + (N - 1)b_G, \quad g_D = a_G. \end{aligned} \quad (15)$$

Since relative treatment sums to zero, it also has no effect on the equally weighted average of the bilateral exchange rates, which is the first equality in (13).

Proposition 2.3 is model-free: it says only that exchangeability reduces the adjustment problem to two scalars, without saying what those scalars are. Section 4 computes them for the nested-CES model.

3 The Nested-CES Model

We now specify the N -country model whose matrices J and G the rest of the paper characterizes.

3.1 Production and preferences

Countries are indexed by $i, j \in \{1, \dots, N\}$. Where useful, we use A for the tariffing country, B for the tariff target, and C for an untariffed bystander. Each country produces a country-specific tradable and a non-tradable from labor:

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i,$$

with perfect competition and the producer-price normalization $P_{T_i} = 1$ in i 's currency, so

$$w_i = A_{T_i}, \quad P_{N_i} = A_{T_i} / A_{N_i}.$$

Wages are constant in own currency, so relative adjustment occurs through exchange rates.

Preferences are nested CES:

$$\begin{aligned} C_i &= C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} && \text{(tradable share } \alpha_T), \\ C_{T_i} &= \left[\alpha_D^{1/\eta} C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} && \text{(home weight } \alpha_D, \text{ elasticity } \eta), \\ M_i &= \left[\sum_{j \neq i} \beta_{ij}^{1/\rho} C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} && \text{(origin weights } \beta_{ij}, \text{ elasticity } \rho). \end{aligned}$$

The nesting is chosen precisely to separate the two substitution margins that the adjustment problem turns on: the elasticity η governs substitution between domestic and imported tradables, while ρ governs substitution across foreign origins conditional on importing. The latter margin is inactive at $N = 2$ and active at $N \geq 3$.

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N-1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Home bias remains through α_D . Full symmetry between domestic and foreign tradables would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, which we do not impose.

3.2 Tariffs, prices, income, and trade balance

Tariffs $\tau_{ij} \geq 0$ are ad valorem and are levied by i on imports from j , with $\tau_{ii} = 0$. Consumer prices in i 's currency are $p_{ij} = (1 + \tau_{ij})e_{ij}$ and $p_{ii} = 1$, where the latter follows from the normalization of the domestic tradable price. The corresponding price indices are

$$P_{M_i} = \left[\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1-\alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad P_i = \kappa_i P_{T_i}^{\alpha_T} P_{N_i}^{1-\alpha_T}. \quad (16)$$

Expenditure shares of income are

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad s_{ij} = \alpha_T (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i, \quad (17)$$

so $s_{ij}I_i$ is tariff-inclusive expenditure at consumer prices and $s_{ij}I_i/(1 + \tau_{ij})$ is the tariff-exclusive border value. Tariff revenue is rebated lump-sum:

$$I_i = w_i L_i + \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij} I_i \implies I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (18)$$

Equation (18) determines income in i 's own currency. To aggregate bilateral flows across countries, take A 's currency as the numeraire and define common-currency income as $\tilde{I}_i \equiv e_{Ai} I_i$. Expenditure shares are unchanged by this conversion. Since tariff revenue is a domestic transfer, trade balances are valued at tariff-exclusive border prices in the common currency:

$$\text{TB}_i = \sum_{k \neq i} f_{ki} - \sum_{j \neq i} f_{ij}, \quad f_{ij} \equiv \frac{s_{ij} \tilde{I}_i}{1 + \tau_{ij}}, \quad (19)$$

where f_{ij} is the border value, in A 's currency, of i 's imports from j .

By Walras' law, only $N - 1$ trade-balance conditions are independent. We assume no internationally traded assets, so $\text{TB}_i = 0$ for all i . Given $\{\tau_{ij}\}$, the $N - 1$ independent exchange rates therefore solve $\text{TB}_i = 0$ for $i \neq A$.

3.3 Exchange-rate concepts

Define e_{ij} as units of i 's currency per unit of j 's, with $e_{ii} = 1$ and $e_{ij} = e_{ik} e_{kj}$. A rise in e_{ij} is therefore a depreciation of i against j . Log changes in bilateral exchange rates relative to A are denoted by $\nu_j \equiv d \log e_{Aj}$, with $\nu_A = 0$.

Terms of trade and real exchange rates are

$$\text{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij} P_{T_j}} = \frac{1}{e_{ij}}, \quad q_{ij} \equiv \frac{e_{ij} P_j}{P_i}. \quad (20)$$

Nominal and real effective exchange-rate changes are

$$d\nu_i^E \equiv \sum_{j \neq i} w_{ij} d \log e_{ij}, \quad dq_i^E \equiv \sum_{j \neq i} w_{ij} d \log q_{ij}, \quad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})}, \quad (21)$$

with $w_{ij} = 1/(N - 1)$ at the symmetric benchmark. Under the normalization of domestic tradable producer prices, $d \log e_{ij} = -d \log \text{ToT}_{ij}$, so the nominal exchange-rate results can equivalently be expressed in terms of the terms of trade.

4 Tariffs and Exchange Rates in the Nested-CES Model

We now specialize the general results of Section 2 to the nested-CES model. Section 4.1 identifies the generic bilateral coefficients of Section 2.1 with model objects; Section 4.2 gives the N -country characterization; Sections 4.3–4.4 then draw out the economics.

4.1 The two-country case

The general linearization for N countries is given in Appendix B; here we need only its $N = 2$ case. Write $m \equiv \alpha_T(1 - \alpha_D)$ for the import share of income, so that $s_{AB} = s_{BA} = m$ at the balanced free-trade baseline with normalized incomes, and let $\nu_B \equiv d \log e_{AB}$. Suppose A imposes $d\tau = dt_{AB}$. Each country has a single import origin, so $dp_{AB} = d \log P_{M_A} = \nu_B + d\tau$ and $dp_{BA} = d \log P_{M_B} = -\nu_B$, and the cross-origin margin in (17) is inactive. Log-differentiating (17) and (18), with incomes in the common currency,

$$\begin{aligned} d \log s_{AB} &= (1 - \eta)\alpha_D(\nu_B + d\tau), & d \log s_{BA} &= -(1 - \eta)\alpha_D \nu_B, \\ d \log \tilde{I}_A &= m d\tau, & d \log \tilde{I}_B &= \nu_B. \end{aligned}$$

Substituting into (19) and using that the tariff wedge lowers the border value of A 's imports by $d\tau$,

$$dTB_A = m \left[(1 + 2\alpha_D(\eta - 1))\nu_B + (1 - m + \alpha_D(\eta - 1))d\tau \right].$$

Define

$$\rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D) = \alpha_D \eta + (1 - \alpha_D)(1 - \alpha_T) > 0. \quad (22)$$

The equilibrium condition can then be written as

$$dTB_A = \underbrace{m D_2 \nu_B}_{\text{exchange-rate effect}} + \underbrace{m \rho_1^* d\tau}_{\text{tariff effect at fixed exchange rates}} = 0, \quad D_2 \equiv 2\alpha_D(\eta - 1) + 1, \quad (23)$$

and therefore

$$\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = - \frac{\rho_1^*}{D_2} \quad (24)$$

Comparing (23) with (1), the generic bilateral coefficients of Section 2.1 are $TB_{A\nu} = mD_2$ and $TB_{A\tau} = m\rho_1^*$, and (24) is (2). In the notation of (3), $-J = mD_2$ and $F = -m\rho_1^*$.

The coefficient ρ_1^* summarizes the trade-balance effect of the tariff at unchanged exchange rates. It reflects substitution between domestic and imported goods, governed by η , together with the associated income and expenditure effects. The second form in (22) separates the two: $\alpha_D \eta$ is switching toward the home tradable, and $(1 - \alpha_D)(1 - \alpha_T)$ is expenditure absorption outside imported tradables. The elasticity ρ , which governs substitution across foreign import origins, is inactive with only two countries. It becomes relevant for $N \geq 3$, when a tariff on one partner can redirect import demand toward other foreign suppliers.

In reduced form, let export volume be $X(1/e)$ and import volume be $M((1 + \tau)e)$, with elasticities ε_X and ε_M . Balanced trade gives $\partial TB_A / \partial e = M(\varepsilon_X + \varepsilon_M - 1)$. Comparing this expression

with (23) per unit of import value gives

$$\varepsilon_X + \varepsilon_M - 1 = D_2, \quad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1.$$

An appreciation therefore restores trade balance after a tariff if and only if the textbook Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$ holds. In our nested-CES model this condition is $D_2 > 0$, which holds whenever $\eta \geq 1$. So D_2 is the object that decides whether exchange-rate adjustment is stabilizing, and ρ_1^* the object that measures the disturbance it has to absorb. Appendix A gives the exact finite-tariff Cobb–Douglas solution.

4.2 The N -country characterization

At the symmetric benchmark A 's partners are symmetric in the sense of Section 2.4, so Proposition 2.3 applies and the entire adjustment problem reduces to the two pairs (j_S, j_D) and (g_S, g_D) . The following theorem evaluates them in the nested-CES model; Appendix B gives the linearized system from which they are obtained.

Theorem 4.1 (Tariffs and exchange rates at the symmetric benchmark). *Consider the nested-CES model of Section 3 at the symmetric benchmark, with $N \geq 3$, and let $t = (t_j)_{j \neq A}$ be any vector of tariffs imposed by A , decomposed as in (11). Define*

$$D_N \equiv N\alpha_D(\eta - 1) + (N - 2)\rho + 1. \quad (25)$$

Then MML (6) holds if and only if $D_N > 0$, for which $\eta \geq 1$ is sufficient. If $D_N > 0$, the coefficients of Proposition 2.3 are

$$\lambda_S = \frac{g_S}{j_S} = -\frac{(N - 1)\rho_1^*}{D_N}, \quad \lambda_D = \frac{g_D}{j_D} = -\frac{(N - 1)\rho}{ND_N}, \quad (26)$$

both strictly negative, and the equilibrium exchange-rate response to t is

$$\nu_j = \lambda_S \bar{t} + \lambda_D (t_j - \bar{t}), \quad (27)$$

with $d\nu_A^E = \lambda_S \bar{t}$, $\nu_j - d\nu_A^E = \lambda_D \tilde{t}_j$ and $d \log e_{jk} = \lambda_D (t_k - t_j)$ as in (13).

Thus D_N is the N -country counterpart of the two-country adjustment coefficient $D_2 = \varepsilon_X + \varepsilon_M - 1$. Average protection determines the effective exchange-rate response, while relative treatment determines deviations of bilateral rates from that common response. In particular, changes in relative treatment that leave $\bar{t}$ unchanged do not affect A 's effective exchange rate.

A useful restriction follows when the two substitution elasticities coincide.

Corollary 4.2 (Single elasticity). *If $\rho = \eta = \sigma$ and $\alpha_D \geq 1/N$, then*

$$N\rho_1^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0,$$

so $\rho \leq N\rho_1^*$ and hence $|\lambda_D| \leq |\lambda_S|$. Without imposing $\rho = \eta$, the same restriction gives

$$N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0,$$

so $\rho > \eta$ is necessary, but not sufficient, for $\rho > N\rho_1^*$.

Equality in the first expression requires $\alpha_D = 1/N$ and $\alpha_T = 1$. Section 4.4 gives the bilateral response that these inequalities govern, and Appendix C the corresponding real exchange-rate results.

4.3 Adjustment coefficients and parameters

Theorem 4.1 rests on two pairs of eigenvalues. For the tariff disturbance,

$$g_S = -\frac{m\rho_1^*}{N-1}, \quad g_D = -\frac{m\rho}{N-1}, \quad (28)$$

and for the exchange-rate adjustment,

$$j_S = \frac{mD_N}{(N-1)^2}, \quad j_D = N \frac{mD_N}{(N-1)^2}; \quad (29)$$

Appendix B derives both. Together they account for the roles of the parameters.

The two elasticities enter at distinct margins. The elasticity η governs substitution between domestic and imported tradables and enters only through ρ_1^* and D_N ; it is the sole substitution margin at $N = 2$. The elasticity ρ governs substitution across foreign origins and enters *both* sides of (5): it raises the relative-treatment disturbance g_D in (28) and, through D_N , the adjustment coefficients j_S, j_D in (29). Requiring $D_N > 0$ scales both responses without affecting their signs. The weights α_D and α_T enter only through ρ_1^* and D_N : by (22), α_T appears only in ρ_1^* , through the absorption term $(1 - \alpha_D)(1 - \alpha_T)$, while α_D scales both the home-switching term $\alpha_D\eta$ in ρ_1^* and, through D_N , the strength of exchange-rate adjustment.

The two modes are therefore loaded differently by the tariff and cleared differently by exchange rates. From (28) and (29),

$$\frac{g_D}{g_S} = \frac{\rho}{\rho_1^*} \text{ is independent of } N, \quad \text{while} \quad \frac{j_D}{j_S} = N. \quad (30)$$

The first ratio is the relative loading of the tariff on the relative-treatment and average-protection disturbances; it compares substitution across foreign origins, ρ , with the home-switching and absorption effects collected in ρ_1^* , and it does not depend on N . The second ratio is the relative strength of equilibrium exchange-rate adjustment along the two modes. A common movement $\nu = \bar{\nu} \mathbf{1}$ changes only the $N - 1$ bilateral rates between A and its partners, leaving the cross rates among partners unchanged; a zero-sum movement changes those cross rates as well. Under symmetry the additional margins make a relative movement exactly N times as effective, so a relative

disturbance requires only $1/N$ as much exchange-rate adjustment. Combining the two,

$$\frac{\lambda_D}{\lambda_S} = \frac{g_D}{g_S} \cdot \frac{j_S}{j_D} = \frac{\rho}{N\rho_1^*}. \quad (31)$$

The factor N in (31) therefore comes entirely from the adjustment side. It does *not* arise because trade diversion is diluted across more partners: the impact comparison $g_D/g_S = \rho/\rho_1^*$ is the same at every N .¹

4.4 Application: a unilateral tariff

Suppose A raises its tariff on B alone, so that $t = d\tau \mathbf{e}_B$. Then

$$\bar{t} = \frac{d\tau}{N-1}, \quad \tilde{t}_B = \frac{N-2}{N-1}d\tau, \quad \tilde{t}_C = -\frac{d\tau}{N-1} \quad \text{for each bystander } C.$$

Substituting into (27), A appreciates against the target,

$$\frac{d \log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N} < 0, \quad (32)$$

while the responses against an untargeted bystander are

$$\frac{d \log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{ND_N}, \quad \frac{d \log e_{BC}}{d\tau} = \frac{(N-1)\rho}{ND_N} > 0. \quad (33)$$

Average protection alone carries the effective exchange rate, which appreciates for every $N \geq 2$ and every admissible ρ :

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = -\frac{\rho_1^*}{D_N} < 0. \quad (34)$$

By (33), the bilateral bystander response reverses sign, so that A depreciates against C , if and only if

$$\boxed{\rho > \rho_N^* \equiv N\rho_1^*}. \quad (35)$$

By Corollary 4.2 this reversal is impossible when $\rho = \eta = \sigma$ and $\alpha_D \geq 1/N$. At $\alpha_D = 1/N$ with $\rho = \eta = \sigma$,

$$\frac{d \log e_{AC}}{d\tau} = -\frac{1 - \alpha_T}{N\sigma},$$

which approaches zero from below as $\alpha_T \rightarrow 1$.

This threshold is the payoff of (30). It differs from the condition under which the bystander's trade balance improves at unchanged exchange rates. From (57),

$$F_B = -\frac{m[\rho_1^* + (N-2)\rho]}{(N-1)^2} < 0, \quad F_C = \frac{m(\rho - \rho_1^*)}{(N-1)^2}. \quad (36)$$

Thus $F_C > 0$ when $\rho > \rho_1^*$, whereas the bilateral exchange rate reverses only when $\rho > N\rho_1^*$. The

¹The exact factor N is specific to the symmetric nested-CES benchmark. Appendix D considers asymmetric baselines and Appendix E heterogeneous substitution across origins.

impact threshold is $g_D/g_S = \rho/\rho_1^*$ crossing one, and is independent of N ; the equilibrium threshold is $\lambda_D/\lambda_S = \rho/(N\rho_1^*)$ crossing one, and carries the extra factor $N = j_D/j_S$ from multilateral adjustment. Figure 2 shows the same comparison as a decomposition of the response vector.

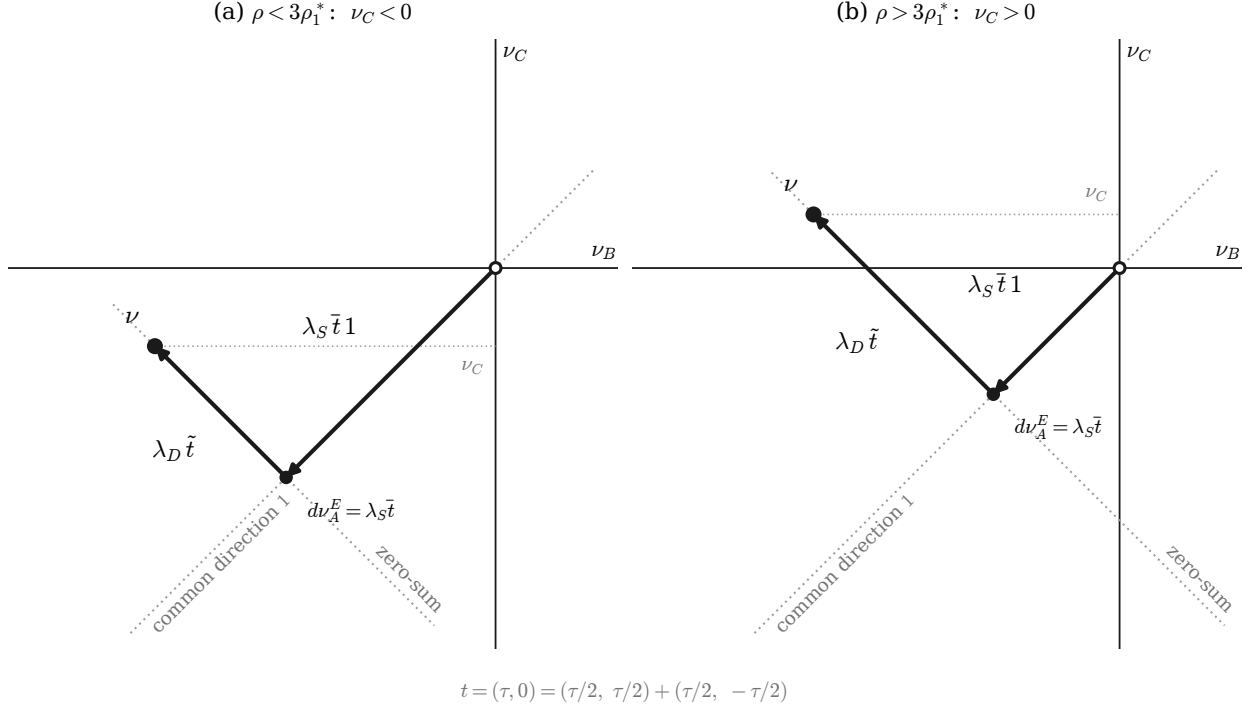


Figure 2: Average protection and relative treatment at $N = 3$. The unilateral tariff $t = (\tau, 0)$ is split as $(\tau/2, \tau/2) + (\tau/2, -\tau/2)$. By Theorem 4.1, the average-protection component moves both bilateral rates equally along the constant direction **1** and carries the whole effective-rate response $dv_A^E = \lambda_S \bar{t}$; the relative-treatment component moves the bilateral rates in opposite directions along the zero-sum direction and leaves the effective rate unchanged. The two components are orthogonal in (ν_B, ν_C) space, so the average-protection point is the projection of ν onto **1**. Panel (b) shows that the bystander rate ν_C turns positive once the relative-treatment step dominates, i.e. once $\rho > 3\rho_1^*$. Drawn at $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$, with $\rho = 2.2$ in panel (a) and $\rho = 5$ in panel (b), so that $\rho_1^* = 1.17$ and $3\rho_1^* = 3.51$.

4.5 Application: multiple targets

Theorem 4.1 also orders bilateral responses directly by tariff treatment. Since $\lambda_D < 0$, (27) gives $\nu_j - \nu_k = \lambda_D(t_j - t_k)$, so with $\bar{t} > 0$, $t_j > t_k$ implies $\nu_j < \nu_k$, and A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_1^* \bar{t} \quad \Longleftrightarrow \quad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_1^*}{\rho}. \quad (37)$$

Thus A appreciates more against partners facing higher tariffs. Depreciation against a relatively lightly taxed partner requires both $\rho > N\rho_1^*$ and a sufficiently large tariff differential.

If A raises its tariff by τ on a set of k partners, $1 \leq k \leq N - 1$, the same decomposition gives,

for every untargeted partner C and every targeted partner B ,

$$\frac{d \log e_{AC}}{\tau} = \frac{k(\rho - N\rho_1^*)}{ND_N}, \quad \frac{d \log e_{AB}}{\tau} = -\frac{k\rho_1^*}{D_N} - \frac{(N-1-k)\rho}{ND_N}, \quad \frac{d\nu_A^E}{\tau} = -\frac{k\rho_1^*}{D_N}. \quad (38)$$

For every $k \leq N-2$, the bystander response has the same threshold $\rho > N\rho_1^*$. When all partners face the same tariff,

$$\frac{d \log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \quad \forall j. \quad (39)$$

All bilateral rates then move together, and substitution across foreign origins plays no role. Appendix B derives (37)–(39).

5 Trade Wars and Preferential Tariffs

The results above also apply when several countries change tariffs simultaneously. Under symmetry, the bilateral exchange-rate effects can be computed for each tariffing country using Theorem 4.1 and then added.

5.1 Trade wars

Bilateral retaliation. Suppose A raises its tariff on B by $d\tau$ and B raises its tariff on A by the same amount. The two effects on the A – B rate cancel, while for any bystander C ,

$$\left. \frac{d \log e_{AB}}{d\tau} \right|^w = 0, \quad \left. \frac{d \log e_{AC}}{d\tau} \right|^w = \frac{\rho - \rho_1^*}{D_N}, \quad \left. \frac{d\nu_A^E}{d\tau} \right|^w = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}. \quad (40)$$

Relative to a unilateral tariff, retaliation lowers the bystander threshold from $N\rho_1^*$ to ρ_1^* . The additional tariff imposed by B affects the A – C rate through relative treatment but not through B 's average protection. Using (26),

$$\left. \frac{d \log e_{AC}}{d\tau} \right|^w = -\frac{\rho_1^*}{D_N} + \frac{\rho}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}. \quad (41)$$

Hence the factor N in the unilateral-tariff threshold disappears under symmetric retaliation.

Global retaliation. Suppose instead that A raises its tariff by $d\tau$ on every partner and each partner raises its tariff by $d\tau$ on A .

Proposition 5.1 (Trade wars). *Under MML:*

- (i) (Bilateral retaliation.) *In a symmetric bilateral trade war between A and B , the belligerents' effective exchange rates depreciate, and every bystander appreciates against both, if and only if $\rho > \rho_1^*$. The threshold is independent of N .*
- (ii) (Global retaliation.) *If A taxes every partner and every partner taxes A , each at rate $d\tau$, then*

$$\frac{d \log e_{Aj}}{d\tau} = \frac{d\nu_A^E}{d\tau} = \frac{(N-2)(\rho - \rho_1^*)}{D_N} \quad \forall j, \quad (42)$$

while all exchange rates among A 's partners are unchanged. Hence A depreciates against every partner if and only if $\rho > \rho_1^*$.

In (ii) the response vanishes at $N = 2$ by symmetry. For $N \geq 3$, the same threshold as in the bilateral trade war applies. Retaliation is a genuinely distinct configuration rather than a relabelling of Section 4.4: because each belligerent's own tariff vector is unilateral, the mirror shock contributes to the A – C rate through relative treatment on both sides, and the two relative-treatment contributions reinforce rather than offset the average-protection term.

5.2 The three-country case

For $N = 3$, the unilateral-tariff system can be written explicitly as

$$J = -\frac{mD_3}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad F = -\frac{m}{4} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \quad D_3 = 3\alpha_D(\eta - 1) + \rho + 1. \quad (43)$$

Hence

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1} F d\tau = \frac{1}{3D_3} \begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau.$$

With $D_3 > 0$, (32)–(33) at $N = 3$ read

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D_3} < 0, \quad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D_3}}, \quad \frac{d \log e_{BC}}{d\tau} = \frac{2\rho}{3D_3} > 0. \quad (44)$$

Thus A appreciates against B , B depreciates against C , and A depreciates against C if and only if $\rho > 3\rho_1^*$.

The three-country case makes the distinction between the fixed-exchange-rate trade-balance effect and the equilibrium exchange-rate response explicit. The bystander's trade balance improves when $\rho > \rho_1^*$, whereas its bilateral exchange rate reverses only when $\rho > 3\rho_1^*$. By (7),

$$\frac{d\nu_C}{d\tau} \propto F_B + 2F_C \propto \rho - 3\rho_1^*.$$

Figure 3 shows the two equilibrium conditions in (ν_B, ν_C) space. Solving (43) row by row, the loci are

$$\text{TB}_B = 0 : \quad \nu_C = 2\nu_B + \frac{\rho + \rho_1^*}{D_3} d\tau, \quad \text{TB}_C = 0 : \quad \nu_C = \frac{1}{2}\nu_B + \frac{\rho - \rho_1^*}{2D_3} d\tau. \quad (45)$$

Both are upward sloping, with $\text{TB}_B = 0$ the steeper. The tariff shifts each locus by an amount proportional to the corresponding impact disturbance in (36): the shift of $\text{TB}_C = 0$ is positive precisely when $\rho > \rho_1^*$. The intermediate case $\rho_1^* < \rho < 3\rho_1^*$ in panel (a) is the one in which these two margins conflict. Holding ν_B at zero, C 's balance clears at a positive $\nu_C = (\rho - \rho_1^*)d\tau/(2D_3)$, so C does benefit from trade diversion on impact. But the much larger shift of $\text{TB}_B = 0$ drives ν_B sharply negative, and because $\text{TB}_C = 0$ has positive slope $1/2$, the induced movement *along* that

locus more than offsets its upward shift. The equilibrium E' therefore lies below the horizontal axis: A still appreciates against C .

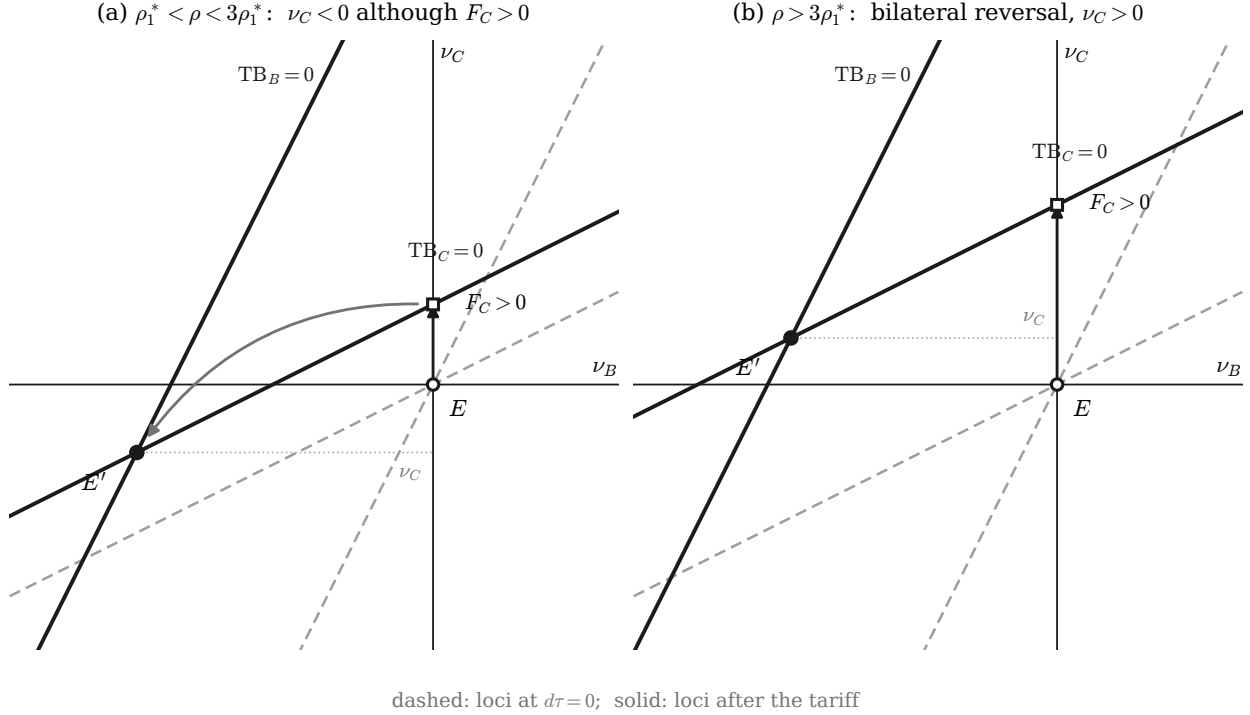


Figure 3: Three-country equilibrium adjustment. The loci are (45); dashed lines are the pre-tariff loci through the initial equilibrium E , solid lines the post-tariff loci intersecting at E' . The open square marks the impact effect on C : the value of ν_C that would clear C 's trade balance at $\nu_B = 0$, which is positive whenever $F_C > 0$, i.e. whenever $\rho > \rho_1^*$. In panel (a), $\rho_1^* < \rho < 3\rho_1^*$: C gains from trade diversion at unchanged exchange rates, but the fall in ν_B along $TB_C = 0$ (curved arrow) more than undoes that gain, so $\nu_C < 0$ in equilibrium. In panel (b), $\rho > 3\rho_1^*$ and the bilateral reversal occurs. Drawn at $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$, with $\rho = 2.2$ in panel (a) and $\rho = 5$ in panel (b), so that $\rho_1^* = 1.17$ and $3\rho_1^* = 3.51$.

5.3 Competing exporters

A standard benchmark in the preferential-trade-agreement literature has one importing country and two competing exporters, following Viner (1950) and Mundell (1964). Importer A is endowed only with the numeraire good y , while exporters B and C are endowed with good x . Preferences are quasilinear, $U_i = u_i(x_i) + y_i$. Country A levies specific tariffs t_B and t_C . Let q_j denote exporter j 's price and p the domestic price in A . Market clearing requires

$$p = q_B + t_B = q_C + t_C, \quad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \quad M'_A < 0, \quad E'_j > 0, \quad (46)$$

where $M_A(p) = x_A^d(p)$ and $E_j(q_j) = \omega_j - x_j^d(q_j)$. Differentiating with respect to t_B gives

$$\frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1), \quad \frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \quad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (47)$$

Proposition 5.2 (Competing-exporter terms-of-trade effect). *A tariff on imports from B worsens B's terms of trade and improves C's terms of trade.*

5.4 Preferential tariffs

The competing-exporter result provides a useful comparison with the three-country case of our model. An improvement in C 's terms of trade corresponds to an increase in e_{AC} by (20). In the competing-exporter model this occurs unconditionally after a tariff on B , whereas in our model (44) requires

$$\rho > 3\rho_1^*.$$

The difference reflects the economic environment. In the competing-exporter model, A does not produce the importable good, so a tariff reallocates demand between the two foreign suppliers without inducing substitution toward a domestically produced import-competing good. In our model, both margins are present: substitution between domestic and imported goods, governed by η , and substitution across foreign import origins, governed by ρ . The terms-of-trade effect on a third country therefore depends on their relative strength.

A preferential tariff reduction gives the corresponding result with the signs reversed. Suppose A lowers its tariff on B alone, so that $t = -\tau \mathbf{e}_B$. For a non-member C , Theorem 4.1 implies that C 's terms of trade worsen if and only if

$$\rho > N\rho_1^*.$$

Thus our model produces the same sign as the standard competing-exporter result only when substitution across foreign origins is sufficiently strong relative to the other general-equilibrium effects.

The two environments are not formally nested. The competing-exporter model has a common importable supplied by B and C and quasilinear preferences, while our model has country-specific differentiated tradables and nested-CES preferences. The comparison is therefore one of mechanisms. The competing-exporter model isolates substitution across foreign suppliers, whereas our framework also includes substitution between domestic and imported goods and the associated income effects.

At $N = 3$, the threshold can be written as

$$\rho_3^* = 3\rho_1^* = 3 + 3\alpha_D(\eta - 1) - 3\alpha_T(1 - \alpha_D). \quad (48)$$

The threshold rises with the domestic-import substitution term and falls with the import share. This helps explain why the third-country terms-of-trade effect is unambiguous in the competing-exporter benchmark but can change sign in the nested-CES model.

Competing-exporter models such as Bagwell and Staiger (1999) and Ornelas (2005) abstract

from trade between the foreign suppliers and use quasilinear preferences, while bloc and endowment models such as Krugman (1991) and Bond and Syropoulos (1996) abstract from domestic production of the importable. Kemp and Wan (1976) consider a more general customs-union environment. The comparison here is not intended to nest these models, but to identify which additional margins determine the sign of third-country terms-of-trade effects in our setting.

6 Conclusion

A tariff affects exchange rates through two steps. At unchanged exchange rates it changes countries' trade balances, summarized by F ; exchange rates then adjust through $(-J)^{-1}$. In the two-country case this reduces to the familiar Marshall–Lerner result: under the elasticity condition, a tariff appreciates the tariffing country's currency.

With three or more countries, a tariff can also redirect expenditure across foreign suppliers. The tariffing country therefore continues to appreciate against the targeted partner, but its exchange rate against an untariffed partner can move in either direction. Under symmetry, the response to any tariff vector separates into an average-protection component and a relative-treatment component. Average protection determines the effective exchange-rate response, while relative treatment determines deviations of bilateral rates from that common response.

For a unilateral tariff on one partner, the tariffing country's effective exchange rate always appreciates under MML. Its bilateral rate against an untariffed bystander reverses only if

$$\rho > N\rho_1^*.$$

The corresponding trade-balance effect on the bystander changes sign already at $\rho > \rho_1^*$, a threshold that is itself independent of N . The difference comes from multilateral exchange-rate adjustment rather than from any dilution of trade diversion: the relative strength of the two tariff disturbances, $g_D/g_S = \rho/\rho_1^*$, does not depend on N , while $j_D/j_S = N$. Under the symmetric nested-CES benchmark, a given relative exchange-rate movement has N times the trade-balance effect of a common movement, so a given relative trade-balance disturbance requires only $1/N$ as much exchange-rate adjustment.

Other tariff configurations alter this comparison. Under symmetric bilateral retaliation, the bystander threshold falls to ρ_1^* and is independent of N . In the standard competing-exporter benchmark, substitution across foreign suppliers generates an unambiguous third-country terms-of-trade effect; in our model, that effect interacts with domestic-import substitution and income effects and therefore need not have an unconditional sign.

The main implication is that the familiar two-country exchange-rate result survives multilateralization most cleanly at the level of the effective exchange rate. Bilateral exchange rates also reflect how trade policy differs across partners, and their responses need not inherit the sign of the aggregate adjustment.

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A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$, expenditure shares are fixed at their weights. From (18), $I_A = w_A L_A / [1 - \frac{\tau}{1+\tau} m]$, and $TB_A = 0$ becomes $e m w_B L_B = m I_A / (1 + \tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1 - m)\tau}, \quad \left. \frac{d \log e}{d\tau} \right|_{\tau=0} = -(1 - m) = -\left. \frac{\rho_1^*}{D_2} \right|_{\eta=1}. \quad (49)$$

Thus $e(\tau)$ is decreasing in τ , so the sign of the linearized response in (24) is exact for all tariff levels in this case. Here $\eta = 1$ fixes the elasticity, not the weight α_D .

B The Linearized System

At the balanced free-trade baseline, let $m \equiv \alpha_T(1 - \alpha_D)$, $b_{ij} \equiv \beta_{ij}$, and $f_{ij} = s_{ij} \tilde{I}_i$, with $\sum_k f_{ki} = \sum_j f_{ij}$. All incomes and flows are measured in the common currency. Linearizing the demand,

income, and trade-balance equations gives

$$dp_{ij} = \nu_j - \nu_i + dt_{ij} \quad (dp_{ii} = 0), \quad (50)$$

$$d \log P_{M_i} = \sum_{j \neq i} b_{ij} dp_{ij}, \quad d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}, \quad (51)$$

$$d \log s_{ii} = -(1 - \eta)(1 - \alpha_D) d \log P_{M_i}, \quad (52)$$

$$d \log s_{ij} = (1 - \rho)(dp_{ij} - d \log P_{M_i}) + (1 - \eta)\alpha_D d \log P_{M_i}, \quad (53)$$

$$d \log \tilde{I}_i = \nu_i + \sum_{j \neq i} s_{ij} dt_{ij}, \quad (54)$$

$$d \text{TB}_i = \sum_{k \neq i} f_{ki} (d \log s_{ki} + d \log \tilde{I}_k - dt_{ki}) - \sum_{j \neq i} f_{ij} (d \log s_{ij} + d \log \tilde{I}_i - dt_{ij}). \quad (55)$$

These equations determine the model-specific matrices J and G in (3). At $N = 2$ they reduce to the calculation in Section 4.1.

At the symmetric baseline, every off-diagonal element of the full $N \times N$ exchange-rate Jacobian is

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad D_N \equiv N\alpha_D(\eta - 1) + (N-2)\rho + 1. \quad (56)$$

Lemma B.1 below gives the derivation. MML holds if and only if $D_N > 0$. In particular,

$$\eta \geq 1 \implies D_N > 0 \iff \text{MML (6) under symmetry.}$$

Thus D_N is the N -country counterpart of the two-country adjustment coefficient $D_2 = \varepsilon_X + \varepsilon_M - 1$.

Lemma B.1 (Full Jacobian at the symmetric baseline). *At the symmetric balanced baseline of Appendix B, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full $N \times N$ Jacobian of the nested-CES model is*

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad \tilde{J}_{ii} = -\frac{m}{N-1} D_N, \quad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1.$$

Proof. Set $dt_{jk} = 0$, so $dp_{ij} = \nu_j - \nu_i$ and $d \log \tilde{I}_i = \nu_i$. Writing $S \equiv \sum_j \nu_j$, (51) gives

$$d \log P_{M_k} = \frac{S - N\nu_k}{N-1}, \quad \frac{\partial d \log P_{M_k}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side,

$$\sum_{j \neq i} (dp_{ij} - d \log P_{M_i}) = 0,$$

so the $(1 - \rho)$ terms cancel when (53) is summed over $j \neq i$. Thus,

$$\sum_{j \neq i} d \log s_{ij} = (N-1)(1 - \eta)\alpha_D d \log P_{M_i}.$$

Substituting into (55) gives

$$dTB_i = \frac{m}{N-1} \left[\sum_{k \neq i} (d \log s_{ki} + \nu_k) - (N-1)(1-\eta)\alpha_D d \log P_{M_i} - (N-1)\nu_i \right].$$

Fix $l \neq i$ and differentiate term by term:

$$\frac{\partial}{\partial \nu_l} \sum_{k \neq i} d \log s_{ki} = -(1-\rho) \frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import-side price-index term contributes $-(1-\eta)\alpha_D$, and the import income term contributes zero. Collecting terms gives

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} \left[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1) \right] = \frac{m}{(N-1)^2} D_N.$$

The diagonal follows from (8):

$$\tilde{J}_{ii} = -(N-1)\tilde{J}_{il} = -\frac{m}{N-1} D_N.$$

At $N = 2$ this gives $\tilde{J}_{AB} = mD_2$, the coefficient in (23); deleting row and column A at $N = 3$ reproduces J in (43). $\square$

The two-country elasticity form. The linearization in Section 4.1 gives (23), with $D_2 = 1 + 2\alpha_D(\eta-1)$ and $\rho_1^* = 1 - m + \alpha_D(\eta-1)$, and (24) follows. For the elasticity form, hold income fixed and let $M(p)$ denote import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB}I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1 - (1-\eta)\alpha_D = 1 + \alpha_D(\eta-1).$$

By symmetry, $\varepsilon_X = 1 + \alpha_D(\eta-1)$. Differentiating $TB_A(e, \tau) = X(1/e) - eM((1+\tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives

$$\frac{\partial TB_A}{\partial e} = M(\varepsilon_X + \varepsilon_M - 1).$$

Hence $\varepsilon_X + \varepsilon_M - 1 = 2\alpha_D(\eta-1) + 1 = D_2$, so the stability condition $D_2 > 0$ is exactly $\varepsilon_X + \varepsilon_M > 1$. Since $\rho_1^* > 0$ by (22), the slope $-\rho_1^*/D_2$ is negative whenever $D_2 > 0$.

The tariff disturbance and the four eigenvalues. At unchanged exchange rates, the effect on partner i 's trade balance follows from (55). Since only i 's exports to A are directly affected,

$$dTB_i|_{\nu \text{ fixed}} = f_{Ai}(d \log s_{Ai} + d \log \tilde{I}_A - t_i) = \frac{m}{N-1} \left[-\rho_1^* \bar{t} - \rho(t_i - \bar{t}) \right]. \quad (57)$$

Reading off the coefficients on $\bar{t}$ and $\tilde{t}_i$ gives the tariff-effect eigenvalues (28). Deleting row and column A from (56) gives the exchange-rate adjustment matrix

$$-J = c(NI - \mathbf{1}\mathbf{1}'), \quad c \equiv \frac{mD_N}{(N-1)^2}, \quad (58)$$

whose eigenvalues are $j_S = c$ on $\text{span}\{\mathbf{1}\}$ and $j_D = Nc$ on the zero-sum subspace, which is (29). Dividing (28) by (29) gives (26).

Direct derivation of (32). The unilateral slopes can also be obtained by inverting a reduced two-equation system. Since the equilibrium is unique under MML and the system is invariant to permutations of the $N - 2$ bystanders, all bystanders have the same response ν_C . The equilibrium therefore reduces to the conditions for B and one representative bystander.

Equations (50)–(51) give

$$d \log P_{M_A} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \quad d \log P_{M_B} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \quad d \log P_{M_C} = \frac{\nu_B - 2\nu_C}{N-1}. \quad (59)$$

Also,

$$d \log \tilde{I}_A = \frac{m}{N-1} d\tau, \quad d \log \tilde{I}_B = \nu_B, \quad d \log \tilde{I}_C = \nu_C.$$

The deviations of bilateral prices from the corresponding import indices are

$$\begin{aligned} dp_{AB} - d \log P_{M_A} &= \frac{N-2}{N-1}(\nu_B + d\tau - \nu_C), & dp_{AC} - d \log P_{M_A} &= \frac{1}{N-1}(\nu_C - \nu_B - d\tau), \\ dp_{BA} - d \log P_{M_B} &= -\frac{N-2}{N-1}\nu_C, & dp_{BC} - d \log P_{M_B} &= \frac{1}{N-1}\nu_C, \\ dp_{CA} - d \log P_{M_C} &= -\frac{1}{N-1}[(N-3)\nu_C + \nu_B], & dp_{CC'} - d \log P_{M_C} &= \frac{1}{N-1}(2\nu_C - \nu_B), \\ dp_{CB} - d \log P_{M_C} &= \frac{1}{N-1}[(N-2)\nu_B - (N-3)\nu_C]. \end{aligned}$$

Using (53) and flow values $m/(N-1)$, collecting the two trade-balance conditions gives

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \quad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1. \quad (60)$$

The matrix has determinant N , so inversion gives (32). At $N = 3$ this reduces to (44), and at $N = 2$ to (24).

Derivation of (32)–(35). By Theorem 4.1,

$$\nu = \lambda_S \bar{t} \mathbf{1} + \lambda_D \tilde{t}.$$

For $t = d\tau \mathbf{e}_B$,

$$\bar{t} = \frac{d\tau}{N-1}, \quad \tilde{t}_C = -\frac{d\tau}{N-1},$$

so

$$\nu_C = \frac{\lambda_S - \lambda_D}{N-1} d\tau = \frac{\rho - N\rho_1^*}{ND_N} d\tau.$$

Under MML, $D_N > 0$, so

$$\text{sign} \frac{d \log e_{AC}}{d\tau} = \text{sign}(\rho - N\rho_1^*).$$

The same substitution with

$$\tilde{t}_B = \frac{N-2}{N-1} d\tau$$

gives the remaining expressions in (32).

Derivation of (34). By Theorem 4.1,

$$d\nu_A^E = \lambda_S \bar{t}$$

with symmetric weights $w_{Aj} = 1/(N-1)$. For $t = d\tau \mathbf{e}_B$,

$$\bar{t} = \frac{d\tau}{N-1},$$

so

$$\frac{d\nu_A^E}{d\tau} = \frac{\lambda_S}{N-1} = -\frac{\rho_1^*}{D_N}.$$

By (22), $\rho_1^* > 0$, while $D_N > 0$ under MML. The response is therefore negative for every $N \geq 2$ and every admissible ρ .

Derivation of (37). From (27),

$$\nu_j - \nu_k = \lambda_D(t_j - t_k),$$

and $\lambda_D < 0$, proving the ordering result. Moreover,

$$\nu_j > 0$$

if and only if

$$\lambda_S \bar{t} + \lambda_D(t_j - \bar{t}) > 0.$$

Substituting (26) and dividing by $(N-1)/D_N > 0$ gives

$$-\rho_1^* \bar{t} - \frac{\rho}{N}(t_j - \bar{t}) > 0,$$

or

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_1^* \bar{t}.$$

For $\bar{t} > 0$, this is equivalent to

$$\frac{t_j}{\bar{t}} < 1 - \frac{N\rho_1^*}{\rho}.$$

The right-hand side is positive only if $\rho > N\rho_1^*$.

Derivation of (38) and (39). With $t_j = \tau$ on the k targets and zero otherwise,

$$\bar{t} = \frac{k\tau}{N-1}, \quad \tilde{t}_B = \frac{N-1-k}{N-1}\tau$$

for a target, and

$$\tilde{t}_C = -\frac{k\tau}{N-1}$$

for a bystander. Substituting into (27) gives

$$\nu_C = \frac{k\tau}{N-1}(\lambda_S - \lambda_D) = \frac{k\tau}{ND_N}(\rho - N\rho_1^*),$$

together with the target and NEER responses in (38). For every $k \leq N-2$, the sign of the bystander response is therefore determined by $\rho - N\rho_1^*$. At $k = N-1$ there is no bystander and $\tilde{t} = 0$, giving (39).

The three-country system. At $N = 3$, (50)–(51) give

$$d \log P_{MA} = \frac{1}{2}(\nu_B + \nu_C + d\tau), \quad d \log P_{MB} = \frac{1}{2}\nu_C - \nu_B, \quad d \log P_{MC} = \frac{1}{2}\nu_B - \nu_C.$$

Also,

$$d \log \tilde{I}_A = \frac{m}{2}d\tau, \quad d \log \tilde{I}_B = \nu_B, \quad d \log \tilde{I}_C = \nu_C.$$

The deviations of bilateral prices from the corresponding import indices are

$$\begin{aligned} dp_{AB} - d \log P_{MA} &= \frac{1}{2}(\nu_B - \nu_C + d\tau), & dp_{AC} - d \log P_{MA} &= \frac{1}{2}(\nu_C - \nu_B - d\tau), \\ dp_{BA} - d \log P_{MB} &= -\frac{1}{2}\nu_C, & dp_{BC} - d \log P_{MB} &= \frac{1}{2}\nu_C, \\ dp_{CA} - d \log P_{MC} &= -\frac{1}{2}\nu_B, & dp_{CB} - d \log P_{MC} &= \frac{1}{2}\nu_B. \end{aligned}$$

The share changes then follow from (53). With flow values $m/2$, (55) gives

$$\begin{aligned} \frac{2}{m} d \text{TB}_B &= (d \log s_{AB} + \frac{m}{2}d\tau - d\tau) + (d \log s_{CB} + \nu_C) - (d \log s_{BA} + \nu_B) - (d \log s_{BC} + \nu_B), \\ \frac{2}{m} d \text{TB}_C &= (d \log s_{AC} + \frac{m}{2}d\tau) + (d \log s_{BC} + \nu_B) - (d \log s_{CA} + \nu_C) - (d \log s_{CB} + \nu_C). \end{aligned}$$

For example, the coefficient on ν_B in $\frac{2}{m}d \text{TB}_B$ is

$$\frac{1}{2}[(1-\rho) + (1-\eta)\alpha_D] + \frac{1}{2}[(1-\rho) + (1-\eta)\alpha_D] + 2(1-\eta)\alpha_D - 2 = -D_3.$$

Collecting the remaining coefficients gives (43), and inverting gives (44). With $D_3 > 0$, the numerator for e_{AB} is negative and the numerator for e_{BC} is positive. The bystander response satisfies

$$\text{sign} \frac{d \log e_{AC}}{d\tau} = \text{sign}(\rho - 3\rho_1^*).$$

C Real exchange rates

This appendix gives the real exchange-rate counterparts of the results in Sections 4.2–4.4.

From (20) and (16),

$$d \log q_{Aj} = \nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}).$$

For a tariff vector t , symmetry implies

$$d \log P_{M_A} = d\nu_A^E + \bar{t}, \quad d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} \nu_j$$

for a partner that imposes no tariff. Hence

$$\begin{aligned} d \log q_{Aj} &= \left(1 - \frac{mN}{N-1}\right) \nu_j - m \bar{t}, \\ dq_A^E &= \left[\left(1 - \frac{mN}{N-1}\right) \lambda_S - m\right] \bar{t}, \\ d \log q_{Aj} - dq_A^E &= \left(1 - \frac{mN}{N-1}\right) \lambda_D \tilde{t}_j. \end{aligned} \tag{61}$$

Thus average protection determines the real effective exchange-rate response, while relative treatment determines bilateral deviations around it, subject to the price-index adjustment in (61).

For a unilateral tariff, $\bar{t} = d\tau/(N-1)$ and

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) \nu_j - \frac{m}{N-1} d\tau, \quad j = B, C. \tag{62}$$

Setting $d \log q_{AC} = 0$ and using (32), for $m < 1/N$,

$$\rho_q^*(N) = \frac{[(N-1) - mN] N \rho_1^* + mN [N \alpha_D (\eta - 1) + 1]}{(N-1)(1 - mN)} > N \rho_1^*. \tag{63}$$

Averaging (62) gives

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_1^*}{D_N} - \frac{m}{N-1} < 0 \quad \text{for all } \rho, N \left(m < \frac{N-1}{N}\right). \tag{64}$$

Proposition C.1 (Real exchange rates). *Under MML, (i) the real bilateral reversal threshold exceeds the nominal threshold, $\rho_q^*(N) > N \rho_1^*$, and (ii) the REER appreciates for every ρ and N , $dq_A^E/d\tau < 0$.*

Both results reflect the direct price-index effect of the tariff. At a given nominal exchange rate, the tariff raises A 's consumer prices, making a real bilateral depreciation harder to obtain and reinforcing the effective appreciation.

D Asymmetry

The disturbance-adjustment representation (5) and MML (6) do not require symmetry. The closed-form thresholds, the NEER cancellation, and the average-protection/relative-treatment decomposition in Sections 4–5 do. This appendix examines which results survive when partner symmetry is relaxed.

A simple degree argument shows why the single-threshold results need not extend to arbitrary trade networks. The elasticity ρ enters the share responses (53) only through $(1 - \rho)$, so every entry of J and F is linear in ρ . Each entry of $\text{adj}(-J)$ is a determinant of an $(N - 2) \times (N - 2)$ submatrix of $-J$ and is therefore a polynomial of degree at most $N - 2$ in ρ . It follows that each numerator $[\text{adj}(-J)F]_i$ in (5) is a polynomial of degree at most $N - 1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator polynomial. Symmetry reduces the problem to the two invariant components of Section 2.4, yielding the linear thresholds in (35) and (44). At a general asymmetric baseline, the $N = 3$ numerator is already quadratic in ρ , as shown below, and for $N \geq 4$ a response may change sign more than once as ρ varies.

General asymmetric baseline. At any balanced baseline, (50)–(55) continue to hold with trade shares (b_{ij}, h_i) and incomes y_i as coefficients. For $N = 3$, the bystander slope has denominator $\det(-J) > 0$ and a numerator that is quadratic in ρ :

$$\text{sign } \frac{d \log e_{AC}}{d\tau} = \text{sign } (c_2 \rho^2 + c_1 \rho + c_0), \quad c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C. \quad (65)$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. The large- ρ sign is economically relevant only if MML continues to hold over the corresponding parameter range, ensuring $\det(-J) > 0$. Under this condition, $c_2 > 0$ whenever A and C each import from both foreign origins. Hence sufficiently strong substitution across import origins eventually reverses e_{AC} at any such baseline. The relevant threshold is the largest positive root and equals $3\rho_1^*$ at the symmetric point. Away from symmetry, however, there is no universal bilateral threshold.

Structured asymmetry. To examine departures from symmetry while preserving tractability, let the target receive a multiple κ of its symmetric bilateral import share. This parameterization remains comparable across N because the symmetric share falls as $1/(N - 1)$. Assume the weight matrix is symmetric and balanced:

$$b_{AB} = b_{BA} = \frac{\kappa}{N - 1}, \quad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N - 2}, \quad b_{CC'} = \frac{1 - 2b_{CA}}{N - 3}. \quad (66)$$

The bystanders remain interchangeable, so the reduced system again has two unknowns and can be solved in closed form. With NEER weights $w_{Aj} = b_{Aj}$, the response is a rational function of (ρ, N, κ) that reduces to $-\rho_1^*/D_N$ at $\kappa = 1$.

Proposition D.1 (Asymmetric NEER). *Holding κ fixed as $N \rightarrow \infty$,*

$$N \frac{d\nu_A^E}{d\tau} \longrightarrow -\kappa \frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} = -\kappa \lim_{N \rightarrow \infty} \frac{N\rho_1^*}{D_N}. \quad (67)$$

To leading order, asymmetric exposure scales the symmetric NEER response by the relative importance of the target without changing its sign. At finite N , the exact numerator can be written as $-\kappa\rho_1^*$ times a quadratic $Q(\rho)$. If $\kappa \geq 1$, all coefficients of Q are positive for sufficiently large N , so the effective rate appreciates for every ρ . If $\kappa < 1$, Q has a positive root satisfying

$$\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1 - \kappa} (1 + O(N^{-1})),$$

above which the NEER depreciates. When the target receives less than its symmetric trade weight, depreciations against the bystanders can therefore outweigh the appreciation against the target. As $\kappa \rightarrow 0$, $\rho_E^* \rightarrow N\rho_1^*$, so the reversal threshold remains of order N .

Under asymmetry, G and $-J$ generally no longer share the constant and zero-sum eigenspaces of Proposition 2.3. Changes in relative treatment can therefore affect the effective exchange rate. The NEER result is exact under symmetry and approximately robust in the structured asymmetric family above; no corresponding claim is made for arbitrary asymmetric trade networks.

E Heterogeneous bilateral substitution

The benchmark imposes a single elasticity ρ across all foreign origins. This appendix asks what survives when substitution across origins is heterogeneous. We do not embed arbitrary bilateral elasticities in a nested CES aggregator: a CES nest with pair-specific elasticities is not in general an integrable demand system. Instead we work directly with the local representation (3),

$$J d\nu + G dt = 0,$$

and ask what the standard adding-up, symmetry and substitution restrictions on demand deliver for G and J separately. Results established without further assumptions are stated as lemmas and propositions; everything that requires additional structure is flagged as such.

E.1 A local substitution matrix

Retain the outer structure of Section 3: the tradable share α_T , the home weight α_D and the home-versus-imports elasticity η are common across countries, and producer prices are normalized as before. Replace only the CES import aggregator: let each importer i have an arbitrary homothetic import aggregator with unit expenditure function $P_{M_i}(p_i)$ and origin shares $b_{ij} = \partial \log P_{M_i} / \partial \log p_{ij}$, with $\sum_{j \neq i} b_{ij} = 1$. Let ε_{jk}^i denote the compensated elasticity of i 's demand for origin j 's good with

respect to p_{ik} , and collect these in the share-weighted substitution matrix

$$(\mathcal{S}_i)_{jk} \equiv b_{ij} \varepsilon_{jk}^i, \quad j, k \neq i. \quad (68)$$

Assumption 2 (Local substitution structure). *For each i , the matrix $\mathcal{S}_i$ is symmetric, negative semidefinite, satisfies $\mathcal{S}_i \mathbf{1} = 0$, and has nonnegative off-diagonal entries.*

The first three requirements are exactly Slutsky symmetry, concavity of the expenditure function, and homogeneity of degree zero of compensated demand; they are implied by, and impose no more than, homothetic utility maximization. The fourth is gross substitutability across origins, the counterpart of Assumption 1 at the level of a single importer.

Lemma E.1 (Laplacian representation). *$\mathcal{S}_i$ satisfies Assumption 2 if and only if $\mathcal{L}_i \equiv -\mathcal{S}_i$ is a weighted graph Laplacian on i 's set of origins:*

$$\mathcal{L}_i = \sum_{j < k} \sigma_{jk}^i (\mathbf{e}_j - \mathbf{e}_k)(\mathbf{e}_j - \mathbf{e}_k)', \quad \sigma_{jk}^i \equiv \rho_{jk}^i b_{ij} b_{ik} \geq 0. \quad (69)$$

The nested-CES benchmark is the case $\rho_{jk}^i = \rho$ for all (i, j, k) , for which $\mathcal{L}_i = \rho(\text{diag}(b_i) - b_i b_i')$. Under (69) the within-nest share response is

$$b_{ij} d \log b_{ij} = (\mathcal{S}_i dp_i)_j + b_{ij} (dp_{ij} - d \log P_{M_i}), \quad (70)$$

which reduces to $d \log b_{ij} = (1 - \rho)(dp_{ij} - d \log P_{M_i})$ in the benchmark.

Proof. A symmetric matrix with zero row sums and nonpositive off-diagonal entries is, by definition, the Laplacian of the weighted graph whose edge weights are $-(\mathcal{S}_i)_{jk}$; such a matrix is automatically positive semidefinite, since (69) is a nonnegative combination of rank-one positive semidefinite terms. Conversely any $\mathcal{L}_i$ of the form (69) is symmetric, positive semidefinite, annihilates $\mathbf{1}$, and has off-diagonal entries $-\sigma_{jk}^i \leq 0$. Writing $\sigma_{jk}^i = \rho_{jk}^i b_{ij} b_{ik}$ is a normalization, since $b_{ij} > 0$. Equation (70) is the definition of the conditional share $b_{ij} = p_{ij} x_{ij} / (P_{M_i} M_i)$ differentiated at constant M_i , using (68). For $\mathcal{L}_i = \rho(\text{diag}(b_i) - b_i b_i')$, $(\mathcal{S}_i dp_i)_j = -\rho b_{ij} (dp_{ij} - d \log P_{M_i})$, which gives the benchmark expression. $\square$

The coefficient ρ_{jk}^i is the elasticity of substitution between origins j and k in market i . Assumption 2 therefore says exactly this: heterogeneous bilateral substitution is a *weighted network* on origins, one network per importing market, and nothing more.

E.2 Where the disturbance goes

Let A impose tariffs $dt = (dt_{Aj})_{j \neq A}$, let $b_A = (b_{Aj})_{j \neq A}$ be A 's origin shares, and define trade-weighted average protection and relative treatment by

$$\bar{t}_A \equiv b_A' dt, \quad \tilde{t} \equiv dt - \bar{t}_A \mathbf{1}. \quad (71)$$

Proposition E.2 (Tariff disturbance). *Under Assumption 2, at a balanced baseline with no initial tariffs, the trade-balance effects of dt at unchanged exchange rates are, for every $i \neq A$,*

$$d\text{TB}_i|_{\nu \text{ fixed}} = m \tilde{I}_A \left[(\mathcal{S}_A dt)_i - \rho_1^* \bar{t}_A b_{Ai} \right], \quad \text{that is} \quad G = m \tilde{I}_A [\mathcal{S}_A - \rho_1^* b_A b'_A], \quad (72)$$

with ρ_1^* as in (22). Equivalently, using $\mathcal{S}_A \mathbf{1} = 0$,

$$G dt = -m \tilde{I}_A \left[\underbrace{\rho_1^* \bar{t}_A b_A}_{\text{average protection}} + \underbrace{\mathcal{L}_A \tilde{t}}_{\text{relative treatment}} \right]. \quad (73)$$

Proof. Only A 's import prices change, so for $i \neq A$ only the export term in (55) is affected: $d\text{TB}_i = f_{Ai}(d \log s_{Ai} + d \log \tilde{I}_A - dt_{Ai})$ with $f_{Ai} = m \tilde{I}_A b_{Ai}$. At $\nu = 0$, $dp_{Aj} = dt_{Aj}$, so $d \log P_{MA} = b'_A dt = \bar{t}_A$ and, by (70) and the outer nest,

$$b_{Ai} d \log s_{Ai} = (\mathcal{S}_A dt)_i + b_{Ai}(dt_{Ai} - \bar{t}_A) + b_{Ai}(1 - \eta)\alpha_D \bar{t}_A.$$

Tariff revenue gives $d \log \tilde{I}_A = \sum_j s_{Aj} dt_{Aj} = m \bar{t}_A$. Collecting,

$$\frac{d\text{TB}_i}{m \tilde{I}_A} = (\mathcal{S}_A dt)_i - b_{Ai} \bar{t}_A [1 - (1 - \eta)\alpha_D - m] = (\mathcal{S}_A dt)_i - \rho_1^* b_{Ai} \bar{t}_A,$$

since $1 + \alpha_D(\eta - 1) - m = \rho_1^*$ by (22). Writing $\mathcal{S}_A dt = \mathcal{S}_A \tilde{t} = -\mathcal{L}_A \tilde{t}$ gives (73). $\square$

Three consequences are worth recording, all of them general.

(i) The disturbance decomposition survives without exchangeability. Equation (73) is an exact split of $G dt$ into an average-protection term and a relative-treatment term, for any substitution network. What changes relative to the benchmark is that the average is trade-weighted rather than unweighted, and that the relative-treatment disturbance is the vector $\mathcal{L}_A \tilde{t}$ rather than a scalar multiple of $\tilde{t}$.

(ii) Uniform tariffs are unaffected by substitution heterogeneity. Since $\mathcal{L}_A \mathbf{1} = 0$, a uniform tariff $dt = \tau \mathbf{1}$ gives $G \mathbf{1} \tau = -m \tilde{I}_A \rho_1^* \tau b_A$, which does not involve $\{\rho_{jk}^A\}$ at all. This generalizes (39): substitution across foreign origins plays no role when all partners are taxed equally, whatever the pattern of substitution.

(iii) Trade diversion follows the substitution network, not the tariff. Componentwise,

$$(\mathcal{L}_A \tilde{t})_i = \sum_{j \neq i} \sigma_{ij}^A (\tilde{t}_i - \tilde{t}_j), \quad (74)$$

so partner i 's impact gain is large when i is taxed lightly *relative to the origins it competes with most closely in A 's market*, with weights given by the edge weights of A 's substitution network.

Under exchangeability, $\sigma_{ij}^A = \rho/(N-1)^2$ and (74) collapses to $\rho \tilde{t}_i/(N-1)$, recovering (28). A bystander that is a poor substitute for the target receives little diversion however lightly it is taxed.

E.3 How disturbances are cleared

Write $\mathbf{B} \equiv (b_{ij})$ for the $N \times N$ matrix of origin shares, with zero diagonal and unit row sums, $\mathbf{Y} \equiv \text{diag}(\tilde{I}_1, \dots, \tilde{I}_N)$ for common-currency incomes, and

$$\mathbf{K} \equiv \mathbf{B} - \mathbf{I}, \quad \theta \equiv 1 + \alpha_D(\eta - 1) = \varepsilon_M, \quad \hat{\mathcal{S}} \equiv \sum_k m \tilde{I}_k \mathcal{S}_k, \quad (75)$$

where each $\mathcal{S}_k$ is extended to an $N \times N$ matrix by inserting zeros in row and column k . A balanced baseline requires $\sum_k f_{ki} = \sum_j f_{ij}$, that is $\mathbf{K}'\mathbf{y} = 0$ with $\mathbf{y} = (\tilde{I}_i)$.

Proposition E.3 (Exchange-rate Jacobian). *Under Assumption 2 and common $(\alpha_T, \alpha_D, \eta)$, the full $N \times N$ Jacobian at a balanced free-trade baseline is*

$$\tilde{\mathcal{J}} = \hat{\mathcal{S}} - m(\mathbf{Y}\mathbf{K} + \theta \mathbf{K}'\mathbf{Y}\mathbf{K}), \quad \text{equivalently} \quad -\tilde{\mathcal{J}} = \hat{\mathcal{L}} + m(\mathbf{Y}\mathbf{K} + \theta \mathbf{K}'\mathbf{Y}\mathbf{K}), \quad (76)$$

where $\hat{\mathcal{L}} \equiv -\hat{\mathcal{S}}$. Its rows and columns sum to zero, so (8) holds. At the symmetric benchmark $\mathbf{B} = (\mathbf{1}\mathbf{1}' - \mathbf{I})/(N-1)$, $\mathbf{Y} = \mathbf{I}$ and $\rho_{ij}^k = \rho$, (76) reduces to $\tilde{\mathcal{J}}_{il} = mD_N/(N-1)^2$ for $l \neq i$, as in Lemma B.1.

Proof. Set $dt = 0$, so $dp_{ij} = \nu_j - \nu_i$ and $d \log \tilde{I}_i = \nu_i$. Because $\mathcal{S}_i \mathbf{1} = 0$, the own-price level drops out of the substitution term: $(\mathcal{S}_i dp_i)_j = (\mathcal{S}_i \nu)_j$. Write $\bar{\nu}_i \equiv \sum_j b_{ij} \nu_j = (\mathbf{B}\nu)_i$, so $d \log P_{M_i} = \bar{\nu}_i - \nu_i = (\mathbf{K}\nu)_i$.

On i 's import side, summing (70) over j gives $\sum_j b_{ij} d \log b_{ij} = \sum_j db_{ij} = 0$, so only the outer nest survives:

$$\sum_{j \neq i} f_{ij} (d \log s_{ij} + \nu_i) = m \tilde{I}_i \left[(1 - \eta) \alpha_D (\mathbf{K}\nu)_i + \nu_i \right].$$

On the export side, using $dp_{ki} - d \log P_{M_k} = \nu_i - \bar{\nu}_k$,

$$\sum_{k \neq i} f_{ki} (d \log s_{ki} + \nu_k) = \sum_k \left[m \tilde{I}_k (\mathcal{S}_k \nu)_i + f_{ki} (\nu_i - \bar{\nu}_k) + f_{ki} ((1 - \eta) \alpha_D (\mathbf{K}\nu)_k + \nu_k) \right].$$

Balance gives $\sum_k f_{ki} = m \tilde{I}_i$, so the ν_i terms cancel against the import side. Collecting the remainder in matrix form, with $\mathbf{f} = m\mathbf{Y}\mathbf{B}$,

$$\tilde{\mathcal{J}} = \hat{\mathcal{S}} - \mathbf{f}'\mathbf{K} + (1 - \eta) \alpha_D (\mathbf{f}' - m\mathbf{Y})\mathbf{K} = \hat{\mathcal{S}} - m(\mathbf{Y}\mathbf{K} + \mathbf{K}'\mathbf{Y}\mathbf{K}) + (1 - \eta) \alpha_D m\mathbf{K}'\mathbf{Y}\mathbf{K},$$

using $\mathbf{f}' = m\mathbf{B}'\mathbf{Y} = m(\mathbf{I} + \mathbf{K}')\mathbf{Y}$ and $\mathbf{f}' - m\mathbf{Y} = m\mathbf{K}'\mathbf{Y}$. Since $1 - (1 - \eta) \alpha_D = \theta$, this is (76). Row sums vanish because $\mathbf{K}\mathbf{1} = 0$ and $\hat{\mathcal{S}}\mathbf{1} = 0$; column sums vanish because $\mathbf{1}'\hat{\mathcal{S}} = 0$, $\mathbf{y}'\mathbf{K} = (\mathbf{K}'\mathbf{y})' = 0$ by balance, and $\mathbf{1}'\mathbf{K}'\mathbf{Y}\mathbf{K} = (\mathbf{K}\mathbf{1})'\mathbf{Y}\mathbf{K} = 0$. The benchmark reduction is a direct computation; it, and (72) and (76) themselves, are checked against a brute-force linearization in `scripts/verify_heterogeneous_substitution.py`. $\square$

Equation (76) separates the two roles of substitution identified in Section 4.3. The first term is the substitution channel and the remaining terms are the income and expenditure-switching channels:

$$\tilde{J}_{il} = m \left[\underbrace{\sum_{k \neq i, l} \tilde{I}_k \sigma_{il}^k}_{\text{substitution}} - \underbrace{\tilde{I}_i b_{il}}_{\text{income}} - \underbrace{\theta (K' Y K)_{il}}_{\text{expenditure switching}} \right], \quad l \neq i. \quad (77)$$

The substitution component of J is a network Laplacian. By Lemma E.1, $\hat{\mathcal{L}} = -\hat{\mathcal{S}}$ is itself a weighted graph Laplacian on the N countries, with edge weight

$$\hat{\sigma}_{il} = \sum_{k \neq i, l} m \tilde{I}_k \sigma_{il}^k \quad (78)$$

between i and l : the total, summed over third markets and weighted by market size, of how closely i and l compete in those markets. Two countries are strongly linked in the adjustment matrix when their goods are close substitutes *somewhere*, not when they trade with each other. This is the precise sense in which J admits a weighted-network interpretation, and it is exact rather than approximate.

The multilateral Marshall–Lerner condition. Assumption 1 requires $\tilde{J}_{il} \geq 0$ for $l \neq i$. By (77) the substitution term works in favor of it and the income term against it, with the expenditure-switching term ambiguous in general but nonnegative at the benchmark. This is the structural counterpart of $D_N = N\alpha_D(\eta-1) + (N-2)\rho + 1 > 0$: MML can fail only if substitution, across origins and between home and imported goods, is weak relative to the income channel. Heterogeneity per se neither helps nor hurts; what matters is whether the network $\hat{\mathcal{L}}$ is sufficiently strongly weighted and connected.

E.4 What remains of the decomposition

By Proposition E.2 the split of the *disturbance* into average protection and relative treatment is general. The split of the *response* is not: it requires that $(-J)^{-1}$ map each piece back into itself. The following identity isolates what is needed.

Lemma E.4 (When $\mathbf{1}$ is an adjustment eigenvector). *Let J be $\tilde{J}$ with row and column A deleted. Then*

$$J\mathbf{1} = -(\tilde{J}_{iA})_{i \neq A}. \quad (79)$$

Hence $\mathbf{1}$ is an eigenvector of J if and only if $\tilde{J}_{iA}$ takes a common value j_S across partners i , in which case $-J\mathbf{1} = j_S\mathbf{1}$.

Proof. Immediate from the row identity $\sum_l \tilde{J}_{il} = 0$ in (8): for $i \neq A$, $\sum_{l \neq A} \tilde{J}_{il} = -\tilde{J}_{iA}$. $\square$

Condition (79) says that a common movement of A 's currency must affect every partner's trade balance equally. Exchangeability implies it; heterogeneity generally does not. Numerically, for random baselines satisfying Assumption 2, $\mathbf{1}$ is not an eigenvector of J and $-J$ and G do not commute; both properties become exact under exchangeability. Without further structure, therefore:

- the effective exchange rate is not in general determined by average protection alone, and changes in relative treatment that leave $\bar{t}_A$ unchanged can move the NEER;
- there is no single relative-treatment coefficient λ_D .

This is the same failure identified for asymmetric trade weights in Appendix D, arriving now through the substitution structure rather than through the trade shares.

Equivariance and relative-treatment modes. A useful intermediate case is partial symmetry. Suppose a group Γ of permutations of A 's partners leaves the baseline invariant: $PBP' = B$, $PYP' = Y$ and $PS_kP' = S_{P(k)}$ for every $P \in \Gamma$. Then both $-J$ and G are Γ -equivariant, so both preserve each isotypic component of the representation of Γ on $\mathbb{R}^{N-1}$.

Proposition E.5 (Spectral representation under partial symmetry). *Suppose Γ leaves the baseline invariant and let $\mathbb{R}^{N-1} = \bigoplus_r V_r$ be the isotypic decomposition. Then $-J$ and G both preserve each V_r . On any V_r that is one-dimensional, or more generally on which $-J$ acts as a scalar, there is a single response coefficient*

$$\lambda^{(r)} = \frac{g_r}{j_r}, \quad -J|_{V_r} = j_r I, \quad G|_{V_r} = g_r I, \quad (80)$$

and for $t \in V_r$ the response is $\nu = \lambda^{(r)}t$. Exchangeability is the case $\Gamma = \mathcal{S}_{N-1}$, which has exactly two components, $\text{span}\{\mathbf{1}\}$ and $\mathbf{1}^\perp$, recovering Proposition 2.3 with λ_S and λ_D .

Proof. Equivariance gives $P(-J)P' = -J$ and $PGP' = G$ for $P \in \Gamma$, so both commute with the representation and, by Schur's lemma, preserve each isotypic component and act as a scalar on each copy of an irreducible component with multiplicity one. Equation (80) and the response follow. For $\Gamma = \mathcal{S}_{N-1}$ the permutation representation decomposes into the trivial and standard representations, which is (15). $\square$

The single coefficient λ_D therefore becomes a *collection of relative-treatment modes*, one per zero-sum isotypic component, ordered by how strongly the corresponding pattern of relative treatment is resisted by the substitution network.

An illustration: competing blocs. Let A 's $N - 1$ partners split into two blocs, with partners inside a bloc exchangeable and substitution within a bloc stronger than across blocs, so that $\Gamma = \mathcal{S}_k \times \mathcal{S}_{N-1-k}$. Then $\mathbb{R}^{N-1}$ decomposes into a two-dimensional trivial component, spanned by the two bloc indicators, and two zero-sum components, one inside each bloc. Proposition E.5 gives two distinct relative-treatment coefficients $\lambda_D^{(1)}, \lambda_D^{(2)}$: relative treatment *within* a bloc is passed through at a rate that depends on that bloc's internal substitutability. On the two-dimensional trivial component, however, $-J$ and G act as 2×2 matrices that need not commute, so average protection and *between*-bloc relative treatment are mixed: a change in relative treatment across the two blocs generally does move A 's effective exchange rate. In a numerical example with $N = 7$ and blocs of three, the within-bloc modes are exact common eigenvectors of $-J$ and G with distinct coefficients $\lambda_D^{(1)} \neq \lambda_D^{(2)}$, while the two-dimensional trivial component is invariant but not diagonal.

When do J and G share eigenvectors? Collecting the above, the benchmark decomposition extends exactly under any of the following, in decreasing strength:

- (a) Exchangeability of the partners (Proposition 2.3): two modes, λ_S and λ_D .
- (b) Γ -invariance with all zero-sum isotypic components of multiplicity one and $\text{span}\{\mathbf{1}\}$ a component on its own: a common eigenbasis, with λ_S and a spectrum $\{\lambda_D^{(r)}\}$.
- (c) $[-J, G] = 0$ with $\mathbf{1}$ an eigenvector of J : the same conclusion as (b) without any group structure, but this is a restriction jointly on $\mathbf{B}$, Y and $\{\mathcal{S}_k\}$ rather than on the substitution structure alone.

We have not found a condition on $\{\mathcal{S}_k\}$ alone that delivers (c) while leaving $\mathbf{B}$ free, and we conjecture that none exists: by (76) the income and expenditure-switching channels contribute $m(YK + \theta K'YK)$ to $-J$ but nothing to G , so the two matrices can share an eigenbasis only through a coincidence between the trade-share matrix and the substitution networks. Exchangeability buys precisely that coincidence.

Summary. What is general about the benchmark results is: the Laplacian structure of the substitution channel in both G and J (Lemma E.1, Propositions E.2 and E.3); the split of the tariff disturbance into average protection and relative treatment, with a trade-weighted average (Proposition E.2); the irrelevance of substitution structure for uniform tariffs; and the reading of the threshold in Section 4.3, since G and J still respond to ρ -type parameters through separate channels. What exchangeability buys is the reduction of the adjustment operator to two scalars, and with it the exact NEER result and the single coefficient λ_D . The claims in this appendix that rest on additional assumptions are Proposition E.5, which requires a symmetry group, and the conjecture in (c) above.

F Correspondence across N

object	$N = 2$	$N = 3$	general N
adjustment condition	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	MML: $-J$ a nonsingular M-matrix
sufficient primitive condition	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 1)
sign rule	scalar response	weights $\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	$\text{sign}(\sum_j \vartheta_{ij} F_j), \vartheta \geq 0$
adjustment eigenvalues j_S, j_D (29)	$mD_2, -$	$mD_3/4, 3mD_3/4$	$mD_N/(N-1)^2, NmD_N/(N-1)^2$
tariff-effect eigenvalues g_S, g_D (28)	$-m\rho_1^*, -$	$-m\rho_1^*/2, -m\rho/2$	$-m\rho_1^*/(N-1), -m\rho/(N-1)$
disturbance vs. adjustment ratios (30)	$-$	$\frac{g_D}{g_S} = \frac{\rho}{\rho_1^*}, \frac{j_D}{j_S} = 3$	$\frac{g_D}{g_S} = \frac{\rho}{\rho_1^*}$ (free of N), $\frac{j_D}{j_S} = N$
bilateral reversal threshold	none	$3\rho_1^*$	$N\rho_1^*$ under symmetry; polynomial roots in general
real bilateral threshold	none	$\rho_q^*(3)$	$\rho_q^*(N)$ (63)
NEER, unilateral tariff	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$ under symmetry
response coefficients λ_S, λ_D (26)	$-\rho_1^*/D_2, -$	$-2\rho_1^*/D_3, -2\rho/(3D_3)$	$-(N-1)\rho_1^*/D_N, -(N-1)\rho/(ND_N)$
trade-war threshold	$-$	ρ_1^*	ρ_1^* , independent of N

G Proofs

Throughout, $m \equiv \alpha_T(1 - \alpha_D)$, $P \equiv \rho + \alpha_D(\eta - 1)$, and all derivatives are evaluated at the balanced free-trade baseline of Appendix B.

Proof of Lemma 2.2. Scaling every currency value e_{Al} by a common factor leaves all relative prices in (50) unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i , by the same factor. Thus TB_i is homogeneous of degree one in the currency values. By Euler's theorem, $\sum_l \tilde{J}_{il} = TB_i$, which is zero at the balanced baseline, so $\sum_l \tilde{J}_{il} = 0$ for all i . Equivalently, in (55), a common shift in all ν_l affects only the income terms, whose export and import contributions cancel by baseline balance. Summing (19) over i gives $\sum_i TB_i \equiv 0$, so $\sum_i \tilde{J}_{il} = 0$ for all l . Under Assumption 1, $\tilde{J}_{il} \geq 0$ for $l \neq i$, so $\tilde{J}_{ii} = -\sum_{l \neq i} \tilde{J}_{il} \leq 0$. Deleting row and column A then gives (9), with strict inequality in every row for which $\tilde{J}_{iA} > 0$.

Thus $-J$ is a Z-matrix with nonpositive off-diagonal entries, nonnegative diagonal entries, weak diagonal dominance, irreducibility, and strict diagonal dominance in at least one row. Its diagonal

is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$, where $D = \text{diag}(-J) > 0$ and $B \geq 0$. Diagonal dominance implies that every row sum of $D^{-1}B$ is at most one and that at least one is strictly below one. With irreducibility, its spectral radius therefore satisfies $r \equiv \rho(D^{-1}B) < 1$. Hence

$$(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k \geq 0} (D^{-1}B)^k D^{-1} \geq 0.$$

Nonsingularity follows from $r < 1$. Moreover,

$$\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0.$$

Finally, the Gershgorin discs of $-J$ are centered at $|J_{ii}| > 0$ with radii no larger than $|J_{ii}|$. They therefore lie in the closed right half-plane and can intersect the imaginary axis only at the origin, which nonsingularity excludes. Hence $\text{Re } \lambda(-J) > 0$, or equivalently $\text{Re } \lambda(J) < 0$. Given the Z-matrix sign pattern this spectral condition is in fact equivalent to MML, and it is local stability of the tatonnement $\dot{v}_i = k_i \text{TB}_i$ for any $k_i > 0$, so simple dynamic exchange-rate adjustment processes converge to the equilibrium. $\square$

Proof of Proposition 2.3. Let M be an $(N - 1) \times (N - 1)$ matrix with $PMP' = M$ for every permutation matrix P . Conjugating by the transposition of i and j gives $M_{ii} = M_{jj}$, and conjugating by a permutation that sends the ordered pair (i, j) to (k, l) , $i \neq j$, $k \neq l$, gives $M_{ij} = M_{kl}$. Hence

$$M = aI + b\mathbf{1}\mathbf{1}'.$$

Since $\mathbf{1}'\mathbf{1} = N - 1$,

$$M\mathbf{1} = [a + (N - 1)b]\mathbf{1},$$

while $Mv = av$ for every v satisfying $\mathbf{1}'v = 0$. This gives (15) for $-J$ and G .

For $j_S, j_D \neq 0$,

$$(-J)^{-1} = \frac{1}{j_D}I + \frac{1}{N - 1} \left(\frac{1}{j_S} - \frac{1}{j_D} \right) \mathbf{1}\mathbf{1}',$$

as direct multiplication verifies. Thus $(-J)^{-1}$ and G share the constant and zero-sum eigenspaces, with response coefficients g_S/j_S and g_D/j_D . Writing

$$t = \bar{t}\mathbf{1} + \tilde{t}, \quad \mathbf{1}'\tilde{t} = 0,$$

gives

$$\nu = \frac{g_S}{j_S} \bar{t}\mathbf{1} + \frac{g_D}{j_D} \tilde{t},$$

and averaging with equal weights yields

$$d\nu_A^E = \frac{g_S}{j_S} \bar{t}.$$

For (14), the off-diagonal entry of $-J$ is

$$b_J = \frac{j_S - j_D}{N - 1},$$

so $-J$ has the Z-matrix sign pattern if and only if $j_D \geq j_S$. Given $j_D \geq j_S > 0$, the entries of $(-J)^{-1}$ above are nonnegative, so MML holds. Conversely, if $j_S \leq 0$, either $-J$ is singular or its inverse cannot be elementwise nonnegative. At $N = 2$ the zero-sum subspace is trivial and the condition reduces to $j_S > 0$.

Finally, for any two partners $j, k \neq A$,

$$d \log e_{jk} = \nu_k - \nu_j = \lambda_D(\tilde{t}_k - \tilde{t}_j) = \lambda_D(t_k - t_j),$$

which completes (13). $\square$

Proof of Theorem 4.1. At the symmetric benchmark relabeling A 's partners leaves the reduced system unchanged, so Proposition 2.3 applies. Appendix B derives (56), (28) and (29). By (29), $j_S = mD_N/(N-1)^2$ and $j_D = Nj_S$, so $j_D \geq j_S$ always and $j_S > 0$ if and only if $D_N > 0$; by (14), MML is therefore equivalent to $D_N > 0$. If $\eta \geq 1$ then every term of (25) is nonnegative and the last is one, so $D_N > 0$. Given $D_N > 0$, dividing (28) by (29) gives (26),

$$\lambda_S = \frac{g_S}{j_S} = -\frac{m\rho_1^*}{N-1} \cdot \frac{(N-1)^2}{mD_N} = -\frac{(N-1)\rho_1^*}{D_N}, \quad \lambda_D = \frac{g_D}{j_D} = -\frac{(N-1)\rho}{ND_N},$$

both negative since $\rho_1^* > 0$ by (22) and $\rho \geq 0$. Equation (27) and the three statements following it are then (12) and (13). $\square$

Proof of Corollary 4.2. With $\rho = \eta = \sigma$,

$$N\rho_1^* = N[1 + \alpha_D(\sigma - 1) - \alpha_T(1 - \alpha_D)],$$

so

$$N\rho_1^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

For $\alpha_D \geq 1/N$ both terms are nonnegative, with both zero only at $\alpha_D = 1/N$, $\alpha_T = 1$; hence $\rho = \sigma \leq N\rho_1^*$, and by (31) $|\lambda_D| \leq |\lambda_S|$. Since ρ_1^* does not involve ρ ,

$$N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

At $\alpha_D = 1/N$,

$$N\rho_1^* - \sigma = (N-1)(1 - \alpha_T), \quad D_N = (N-1)\sigma,$$

so by (33),

$$\frac{d \log e_{AC}}{d\tau} = -\frac{1 - \alpha_T}{N\sigma}.$$

$\square$

Proof of Proposition 5.1(i). Let primes denote responses to the mirror shock $t_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in (32) gives

$$d \log e'_{BA} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad d \log e'_{BC} = \frac{\rho - N\rho_1^*}{ND_N}.$$

Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$,

$$d \log e'_{AB} = \frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad d \log e'_{AC} = \frac{(N-1)\rho}{ND_N}.$$

By linearity,

$$d \log e_{AB} \Big| ^w = 0,$$

and

$$d \log e_{AC} \Big| ^w = \frac{\rho - N\rho_1^*}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}.$$

By symmetry between A and B , the same expression applies to $d \log e_{BC} \Big| ^w$. Averaging gives

$$\frac{d\nu_A^E}{d\tau} \Big| ^w = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}.$$

Under MML, $D_N > 0$, so the signs are determined by $\rho - \rho_1^*$. □

Proof of Proposition 5.1(ii). Superpose the N tariff vectors. A 's tariff vector is uniform,

$$\bar{t}^{(A)} = d\tau, \quad \tilde{t}^{(A)} = 0,$$

and contributes $\lambda_S d\tau$ to every $d \log e_{Aj}$.

Partner j taxes only A . Its contribution to $d \log e_{Aj}$ is

$$-\left[\lambda_S + (N-2)\lambda_D\right] \frac{d\tau}{N-1},$$

while each of the $N-2$ other partners contributes $-\lambda_D d\tau$. Summing,

$$\begin{aligned} d \log e_{Aj} &= \lambda_S d\tau - \left[\lambda_S + (N-2)\lambda_D\right] \frac{d\tau}{N-1} - (N-2)\lambda_D d\tau \\ &= \frac{(N-2)(\rho - \rho_1^*)}{D_N} d\tau. \end{aligned}$$

The expression is the same for every j , so it also equals the NEER response, and all cross rates among A 's partners are unchanged. □

Proof of Proposition C.1. From (20) and (16), only tradable prices move, so

$$d \log q_{Aj} = \nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}),$$

with

$$d \log P_{T_i}^{\text{own}} = (1 - \alpha_D) d \log P_{M_i}.$$

Using (59),

$$d \log q_{AB} = \left(1 - \frac{mN}{N-1}\right) \nu_B - \frac{m}{N-1} d\tau.$$

The same calculation for $j = C$ gives (62).

For a general tariff vector,

$$d \log P_{M_A} = \frac{1}{N-1} \sum_j (\nu_j + t_j) = d\nu_A^E + \bar{t},$$

while for a partner that imposes no tariff,

$$d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} \nu_j.$$

Substitution gives (61). Setting $d \log q_{AC} = 0$ and substituting ν_C from (32) gives

$$[(N-1) - mN] \frac{\rho - N\rho_1^*}{N} = mD_N(\rho).$$

Solving for ρ gives (63). If $m < 1/N$, the coefficient on the left is positive, implying $\rho_q^*(N) > N\rho_1^*$. Averaging (62) and using (34) gives (64), whose two terms are negative under the stated conditions. $\square$

Proof of Proposition 5.2. The result follows directly from (47). Total differentiation of market clearing gives $dp/dt_B \in (0, 1)$, and $M'_A < 0$ and $E'_j > 0$ then imply the stated signs of dq_B/dt_B and dq_C/dt_B . $\square$

Proof of Proposition D.1. Under the weights in (66), the bystanders remain interchangeable, so the system reduces to two unknowns (ν_B, ν_C) . The corresponding flow values are $m\beta$, $m\gamma$, and $m\xi$, where

$$\beta = \frac{\kappa}{N-1}, \quad \gamma = \frac{1-\beta}{N-2}, \quad \xi = \frac{1-2\gamma}{N-3}.$$

Write

$$\nu_B = \frac{b}{N}, \quad \nu_C = \frac{c}{N},$$

holding $d\tau$ fixed, and expand the two equilibrium conditions in $1/N$ at fixed (κ, ρ) . To leading order,

$$\beta \rightarrow \frac{\kappa}{N}, \quad (N-2)\gamma \rightarrow 1,$$

and

$$d \log P_{M_A} = \frac{\kappa d\tau + c}{N} + O(N^{-2}), \quad d \log P_{M_B} = \frac{c-b}{N} + O(N^{-2}), \quad d \log P_{M_C} = O(N^{-2}).$$

Substituting into (53) and the trade-flow sums gives the limiting system

$$P(b - c) = -\kappa\rho d\tau, \quad P(2c - b) = \kappa(\rho - \rho_1^*) d\tau, \quad P \equiv \rho + \alpha_D(\eta - 1).$$

The limiting matrix

$$P \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

has determinant $P^2 > 0$. Adding the two equations gives

$$c = -\frac{\kappa\rho_1^*}{P} d\tau.$$

Since the NEER weights satisfy $w_{AB} = \beta = O(N^{-1})$ and $(N - 2)w_{AC} \rightarrow 1$,

$$N \frac{d\nu_A^E}{d\tau} \longrightarrow -\frac{\kappa\rho_1^*}{\rho + \alpha_D(\eta - 1)}.$$

Finally,

$$\frac{D_N}{N} \longrightarrow \rho + \alpha_D(\eta - 1),$$

which gives (67). □